

SYSTEMIC RISK AS GEOMETRY: BLIND-SPOT DECOMPOSITION, ADAPTIVE FRICTION, AND APPLICATIONS TO FINANCE, ENERGY, AND NAVIER–STOKES REGULARITY*

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Abstract. We propose a geometric framework for detecting and controlling structural instability in coupled dynamical systems by recasting the problem in terms of energy-topology equivalence. N interacting agents evolve under a potential Φ whose pairwise term encodes contagion; a *Blind-Spot Decomposition Theory* (BSDT) energy functional E_{BS} , calibrated on normal-period data, is shown to be topologically equivalent to Φ via a *Morse-index equivalence* (Theorem C: $\text{ind}_{E_{\text{BS}}}(X_c) = \text{ind}_{\Phi}(X_c)$ at every critical point, with two-sided gradient bounds $c_1 \|\text{grad } \Phi\| \leq \|\text{grad } E_{\text{BS}}\| \leq c_2 \|\text{grad } \Phi\|$ and homotopy equivalence of sublevel sets). This equivalence recasts the instability detection problem as a control problem: a *critical manifold* C^* , characterised by the Morse-index jump $\text{ind} = 0 \rightarrow \text{ind} \geq 1$, separates a stable from an unstable regime. Adaptive friction $\gamma^*(X) = \alpha / \lambda_{\max}(D^2 \Phi_{\text{pair}})$ achieves marginal stability where every fixed coupling fails (Theorem 4.3)—a formal characterisation of procyclicity in financial regulation.

Two physics-based *System Mode engines* operationalise the theory: a Lennard-Jones 6–12 *Molecular Engine* and a Gaussian-attraction / Coulomb-repulsion *Gravity Mode Engine*, unified through a common Lyapunov v2 stabiliser with Armijo backtracking ($c = 10^{-4}$), Input-to-State Stability (ISS) margins, and La Salle convergence certification ($\tau_{\text{LS}} = 10^{-5}$, patience 5). Both engines share a *Fused Topological Scoring* architecture: Morse critical-point alarm, Betti barcode persistence, and a four-operator UDL stack (Phase, Topological, KernelRKHS, Rank) fused at equal weight after min-max normalisation. The BSDT channel weights are determined by a closed-form Fisher variance-ratio statistic (zero label leakage). A *SpectraFalseAlarmFilter* gates scores by topological persistence, and a *FARTargetCalibrator* enforces a target false-alarm rate via piecewise-linear rescaling.

A *Universal Conformal Scoring* layer provides distribution-free anomaly detection: each engine’s physics simulation produces raw scores that are calibrated via split conformal prediction, yielding valid p -values under exchangeability alone. Panel-level aggregation (rolling mean, P90, max, standard deviation, momentum) fuses point-level scores into quarter-level summaries; rank-normalised fusion selects the better of point-level and panel-level views per dataset automatically. On the ERCOT benchmark the Hybrid Engine achieves $\text{AUC} = 0.9997$; on the G-SIB bank-level panel, quarter-level conformal fusion lifts the Gravity Engine from 0.766 to 0.806 (+4.0 pp) while preserving zero COVID false alarms. All reported lead times are measured under a strictly causal expanding-window protocol: each score at quarter t uses only information available up to t , with no future observations, no future labels, and no hindsight re-estimation.

The extended framework is validated on FDIC banking data (6-quarter GFC lead, $\text{AUROC} = 0.867$ with adaptive calibration, zero false alarms at 38.5% recall and 100% precision), the Terra/Luna collapse ($r = 0.996$, +30 h lead), the Texas ERCOT grid (+72 h lead), and three new sensitive-sector benchmarks: FRED macroeconomic indicators ($\text{AUC} = 0.9999$), ERCOT energy-grid dispatch ($\text{AUC} = 0.9940$), and a prospective G-SIB bank-level panel where Mol+ExpoGate first alarms at 2007-Q4—two quarters before the Bear Stearns collapse. The BSDT channels transfer to three-dimensional incompressible Navier–Stokes, where adaptive viscosity $\nu(E_{\text{BS}})$ suppresses peak enstrophy by $1.77\times$ at $\text{Re} \approx 1257$ on a mesh-converged 128^3 pseudo-spectral grid (Taylor–Green vortex; resolution-convergent to within $< 1\%$). A 15-point Reynolds sweep ($\text{Re} = 314\text{--}62832$) reveals a non-monotone profile: suppression peaks at $S_{\max} = 1.43\times$ in the transitional regime ($\text{Re} \approx 628$) and decays as $S(\text{Re}) \approx 1.66 \cdot \text{Re}^{-0.047}$ ($\alpha = -0.047 \pm 0.012$, $p = 0.0016$, $R^2 = 0.55$). All 15 runs complete without blow-up, the BKM integral is finite throughout, and $\gamma_{\max}^* = 0.9959$ uniformly, confirming that the BSDT feedback is active across the full laminar-to-turbulent range.

Key words. systemic risk, Lyapunov stability, entropy production, Onsager gradient flow, phase transition, adaptive friction, Navier–Stokes regularity, enstrophy, blind-spot decomposition, online convex optimisation, physics-based scoring, Lennard-Jones potential, fused topology, false-alarm-rate calibration

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1. Introduction. The 2008 Global Financial Crisis demonstrated that systemic risk resides not in individual institutions but in the geometry of their interactions. Existing frameworks—CoVaR [2], SRISK [7], network contagion models [1, 9]—detect stress *after* it propagates. This paper proposes a geometric framework for detecting and characterising structural instability in coupled systems, and derives adaptive stabilisation as a direct consequence of the underlying energy-topology structure. The key insight is that a detection functional calibrated on normal-period data shares the same Morse-index landscape as the interaction energy—so detecting instability and controlling it become two views of the same geometric object.

Formally, an *equivalence theorem* links two seemingly unrelated objects: (i) a *detection* functional E_{BS} measuring deviation from historical normality, and (ii) the *interaction energy* Φ governing agent dynamics. Theorem C establishes that E_{BS} and Φ share the same Morse-index structure at every critical point, that their sublevel sets are homotopy equivalent, and that gradient norms are equivalent up to computable constants. This identification transforms the instability detection problem into a control problem: damping the system using $\text{grad } E_{BS}$ stabilises it because $\text{grad } E_{BS}$ and $\text{grad } \Phi$ are geometrically aligned. The Morse-index jump from 0 to ≥ 1 provides a topological instability signal that is structurally invariant under perturbation and decoupled from the Lyapunov descent certificate used to guarantee iteration convergence.

A *critical manifold* $\mathcal{C}^*$ separates two dynamical regimes. Below $\mathcal{C}^*$, any fixed coupling suffices for stability. Above $\mathcal{C}^*$, no constant friction stabilises all configurations (Theorem 4.3). The system requires *adaptive* friction $\gamma^*(X)$ that tracks the spectral radius of the pairwise Hessian in real time. This phase-transition structure provides a rigorous dynamical-systems characterisation of procyclicality: time-invariant regulatory rules (such as Basel III’s fixed countercyclical capital buffer) fail *structurally*, not merely in practice.

Contributions.. This paper makes nine contributions:

1. **Blind-Spot Decomposition Theory (BSDT).** Four deviation operators (δ_C : camouflage, δ_G : feature gap, δ_A : activity anomaly, δ_T : temporal novelty) define E_{BS} , a composite energy functional that decomposes structural instability into geometrically interpretable channels, calibrated entirely on normal-period data.
2. **Equivalence Theorem (Theorem C) and Entropy Structure (Theorem 3.13).** Morse-index equivalence $\text{ind}_{E_{BS}} = \text{ind}_{\Phi}$ at every critical point, topological energy equivalence via homotopy of sublevel sets, two-sided gradient bounds, and decoupled stabilisation establish E_{BS} as a Lyapunov function for the potential flow (Proposition 3.8); the system is a generalised Onsager gradient flow with non-negative entropy production $\sigma(X) = \gamma(E_{BS}) \|\text{grad } E_{BS}\|^2 \geq 0$. This equivalence provides the mathematical basis for both detecting instability (via the Morse-index jump) and controlling it (via gradient-aligned damping).
3. **Critical manifold and phase transition (Theorem 4.3).** A spectral threshold determines where constant friction fails and adaptive friction succeeds, providing a rigorous dynamical-systems characterisation of why constant-parameter regulation fails in complex networks.
4. **System Mode engines (Section 6).** A Lennard-Jones Molecular Engine and a Gaussian/Coulomb Gravity Mode Engine share an identical post-simulation operator stack (Phase, Topological, KernelRKHS, Rank) fused via a Morse/Betti/UDL scoring architecture. A Hybrid Engine selects the

- optimal convex blend by cross-validated AUROC maximisation.
5. **Lyapunov v2 convergence guarantee (Proposition 6.1).** Three-part certificate: Armijo–ISS descent, La Salle invariance, and coercivity ensure convergence for all engines with $O(nk)$ per-step complexity.
 6. **Online algorithm with $O(\sqrt{T})$ regret.** The adaptive coupling is implementable via online gradient descent with power-iteration spectral estimates.
 7. **Empirical validation.** On FDIC banking data ($T = 140$ quarters, $N = 7$ institution types, $d = 6$ features), the framework achieves a 6-quarter GFC early warning, $\text{AUROC} \geq 0.70$. On FRED macroeconomic data, $\text{AUC} = 0.9999$; on ERCOT energy-grid dispatch, $\text{AUC} = 0.9940$; on a prospective G-SIB bank-level panel, $z > 2$ six quarters before Lehman.
 8. **Universal conformal scoring (Section 6.6).** Split conformal prediction calibrates each engine’s raw physics scores into valid p -values under exchangeability alone. Panel-level aggregation and rank-normalised fusion automatically select the better of point-level and quarter-level scoring per dataset, improving Gravity AUC on the G-SIB panel by +4.0 pp with no additional hyperparameters.
 9. **Cross-domain universality.** The same geometric framework, without re-training, characterises structural instability in the Terra/Luna collapse ($r = 0.996$), the Texas ERCOT grid failure (+72 h lead), and Navier–Stokes turbulence—suggesting that the energy-topology structure is a general feature of coupled dynamical systems, not specific to any single domain.
 10. **Extension to Navier–Stokes.** The BSDT channel structure transfers to three-dimensional incompressible Navier–Stokes, where adaptive viscosity suppresses enstrophy growth and maintains bounded production/dissipation ratios—a pathway toward regularity results.

Related work. Network-based systemic risk models [1, 9] capture topology but not dynamics. Market-based measures (CoVaR, SRISK) require market prices and operate at the market frequency, missing slow-moving structural drift on regulatory balance sheets. The May–Levin–Sugihara programme [19] imports ecological stability analysis but lacks operational control algorithms. Our framework complements these by providing feedback control with formal stability guarantees. For the Navier–Stokes connection, the Beale–Kato–Majda criterion [4] and Ladyzhenskaya’s classical estimates [15] provide the regularity backdrop.

Outline. Section 2 introduces the agent model and BSDT. Section 3 proves Theorem C. Section 4 establishes the phase transition. Section 5 presents the online algorithm and optimal policy. Section 6 introduces the System Mode engines (Molecular, Gravity, Hybrid) and their shared fused scoring architecture. Section 7 validates on financial data. Section 8 extends to crypto and energy systems. Section 9 transfers the framework to Navier–Stokes. Section 10 concludes.

2. Mathematical Framework.

2.1. Agent Model and GravityEngine Dynamics. Consider N interacting institutions, each described by a state vector $x_i(t) \in \mathbb{R}^d$ of financial characteristics. The joint state $X(t) = (x_1, \dots, x_N) \in \mathbb{R}^{N \times d}$ evolves under the gradient flow

$$\dot{X} = F(X) = -\text{grad } \Phi(X), \quad (2.1)$$

where $\Phi : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}$ decomposes as

$$\Phi(X) = \underbrace{\frac{\alpha}{2} \sum_{i=1}^N \|x_i - \mu\|^2}_{\text{radial spring (mean-reversion)}} + \underbrace{\sum_{i < j} \phi(\|x_i - x_j\|)}_{\text{pairwise interaction}}. \quad (2.2)$$

The pairwise potential $\phi(r) = \gamma \frac{\sigma \sqrt{\pi}}{2} \text{erf}(r/\sigma) - \gamma \lambda \log(r + \varepsilon)$ combines short-range Gaussian attraction with long-range logarithmic repulsion, giving clustering at equilibrium distance $r^* = \sigma \sqrt{\log(1/\lambda)}$.

The *unique Nash equilibrium* under the friction game $\min_{\ell_i} [\text{private cost}_i + \gamma \cdot \text{systemic externality}_i]$ is $\ell_i^*(\gamma) = (\alpha + \gamma w_i)^{-1} r_i$, where w_i is institution i 's network centrality and r_i its return on assets. The optimal coupling γ^* internalises the Pigouvian externality $\tau_i^* = \gamma^* \partial E / \partial \ell_i$, proportional to each institution's marginal contribution to system energy.

2.2. Blind-Spot Decomposition Theory (BSDT). BSDT decomposes systemic risk into four geometrically orthogonal channels of deviation from the normal-period distribution $\mathcal{N}$:

1. **Camouflage** (δ_C): Mahalanobis distance from $\mathcal{N}$. Detects states that appear individually normal but are collectively anomalous: $\delta_C(X) = [(X - \mu_0)^\top \Sigma_0^{-1} (X - \mu_0)]^{1/2}$.
2. **Feature Gap** (δ_G): PCA residual in low-variance directions. Detects motion into historically unoccupied subspaces: $\delta_G(X) = \|(I - P_k)X\|$.
3. **Activity Anomaly** (δ_A): Excess velocity relative to normal-period dynamics: $\delta_A = \max(0, \|\dot{x}_i\| - v_0)$.
4. **Temporal Novelty** (δ_T): Negative log-density under a KDE of historical states: $\delta_T = -\log \hat{p}(x_i(t))$.

The *blind-spot energy functional* is the weighted composite

$$E_{\text{BS}}(X) = \sum_{k=1}^4 w_k \psi_k(\delta_k(X)) + \sum_{i < j} \phi(\|x_i - x_j\|), \quad (2.3)$$

where $\psi_k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are convex, non-decreasing, 1-Lipschitz link functions with $\psi_k(0) = 0$. The pairwise potential in E_{BS} coincides with Φ_{pair} , ensuring structural compatibility between detection and dynamics.

Fisher Variance-Ratio Weighting. The channel weights w_k are determined entirely from the data, requiring no labelled examples. Let $m_k = |\delta_k(X)|$ be the channel magnitude vector across all n observations. We split the data at the 80th and 50th percentiles of total magnitude $M = \sum_k m_k$ into “high” and “low” groups and compute the *Fisher variance ratio*:

$$\text{FR}_k = \frac{(\mu_k^{\text{high}} - \mu_k^{\text{low}})^2}{\text{Var}_k^{\text{high}} + \text{Var}_k^{\text{low}}}, \quad w_k = \frac{\text{FR}_k}{\sum_j \text{FR}_j}. \quad (2.4)$$

This closed-form weighting adaptively assigns importance to whichever channels best separate high-deviation from low-deviation regimes, with zero label leakage by construction.

Sign Discovery and Herding. The sign $\text{sgn}(\mu_k^{\text{high}} - \mu_k^{\text{low}})$ reveals the *direction* of each channel's deviation during stress. On financial data, δ_C receives a *negative* Fisher coefficient—curvature *collapses* during crises, the fingerprint of herding: agents crowd into identical positions, reducing the loss-landscape curvature to near zero. Meanwhile

δ_T (temporal novelty) receives the strongest positive weight, indicating that crises produce genuinely unprecedented states.

The *Multi-Factor Latent Score* (MFLS) is the Frobenius norm $\|\text{grad } E_{\text{BS}}(X_t)\|_F$, serving as a scalar early-warning signal.

3. The Equivalence Theorem. We work with a fixed reference measure $\mathcal{N}$ (the normal-period distribution) and assume Φ satisfies Assumptions 3.1–3.4 below.

3.1. Assumptions. ASSUMPTION 3.1 (Regularity). $\Phi \in C^2(\mathbb{R}^{N \times d})$ is coercive: $\Phi(X) \rightarrow +\infty$ as $\|X\| \rightarrow \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \leq c\}$ is compact, with Lipschitz constant L_c for $\text{grad } \Phi$ on $\mathcal{L}_c$.

ASSUMPTION 3.2 (Strict local minimum). The equilibrium X^* satisfies $\text{grad } \Phi(X^*) = 0$ and $H^* := D^2\Phi(X^*) \succ 0$.

ASSUMPTION 3.3 (BSDT regularity). Each $\delta_i : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^+$ is C^1 and K_i -Lipschitz on $\mathcal{L}_c$. Each $\psi_i : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, with $\psi_i(0) = 0$.

ASSUMPTION 3.4 (Separation).

(S1) **Non-degeneracy.** There exists $\nu > 0$ such that for every $X \in \mathcal{L}_c$ with $\text{grad } \Phi(X) \neq 0$, at least one index $i = i(X) \in \{1, \dots, 4\}$ satisfies $\|\text{grad}_X \delta_i(X)\| \geq \nu \|\text{grad } \Phi(X)\|$.

(S2) **Monotonicity.** For all i and all $X \in \mathcal{L}_c$: $\langle \text{grad}_X \delta_i(X), \text{grad } \Phi(X) \rangle \geq 0$.

ASSUMPTION 3.5 (Morse non-degeneracy). At every critical point X_c of E_{BS} (i.e. $\text{grad } E_{\text{BS}}(X_c) = 0$), the Hessian $D^2 E_{\text{BS}}(X_c)$ is non-degenerate: $\det(D^2 E_{\text{BS}}(X_c)) \neq 0$. Likewise, at every critical point X_c of Φ , $\det(D^2 \Phi(X_c)) \neq 0$.

REMARK 3.6. Morse non-degeneracy is a generic condition: the set of potentials violating it has infinite codimension in the C^2 topology. For the LJ 6–12 and Gaussian/Coulomb potentials ((6.1), (6.3)), non-degeneracy holds at all finite-energy critical points by direct computation of the Hessian eigenvalues.

REMARK 3.7. Part (S2) holds naturally: each δ_i measures deviation from normal-period behaviour, so δ_i increases along directions of increasing instability Φ . For the Mahalanobis channel δ_C , the gradient $\Sigma_0^{-1}(X - \mu_0)/\delta_C$ shares the outward radial direction with $\text{grad } \Phi$. For δ_G , the orthogonal projector $(I - P_k)$ either gives zero or increases in directions exiting the principal subspace. Part (S1) is the existential (rather than universal) requirement that the composite detection gradient is not degenerate.

PROPOSITION 3.8 (Structural derivation of the separation condition). Let $\Sigma_0 \succ 0$ be the Ledoit–Wolf shrinkage estimator with condition number $\kappa(\Sigma_0)$. Then:

- (i) δ_C satisfies (S1) with $\nu_C = \kappa(\Sigma_0)^{-1}[\alpha \sqrt{\lambda_{\max}(\Sigma_0)}]^{-1}$, determined entirely by spectral properties of Σ_0 and the spring constant α . Monotonicity (S2) holds because $\langle \text{grad } \delta_C, \text{grad } \Phi \rangle \geq 0$ in a neighbourhood of X^* (the Mahalanobis and potential gradients share the outward radial direction).
- (ii) δ_G satisfies $\|\text{grad}_X \delta_G\| = 1$ whenever $\delta_G > 0$, giving unconditional (S1). At points with $\delta_G = 0$, the feature gap vanishes and cannot be the active channel; another δ_i provides the bound.
- (iii) δ_A satisfies (S1) with $\nu_A = 2\eta(\lambda_u - \alpha)/L_c$ in the super-critical region.
- (iv) δ_T satisfies (S1) with $\nu_T = h^{-1}/L_c$ outside the kernel bandwidth h .

Since each channel individually satisfies (S1) in a region, and the regions cover $\mathcal{L}_c$, the composite E_{BS} inherits the existential bound with $\nu \geq \max(\nu_C, \nu_G, \nu_A, \nu_T)$ at each X .

Proof. Part (i): $\text{grad}_X \delta_C = \Sigma_0^{-1}(X - \mu_0)/\delta_C(X)$; by the Rayleigh quotient, $\|\Sigma_0^{-1}v\|^2 \geq \lambda_{\max}(\Sigma_0)^{-2}\|v\|^2$, and $\delta_C(X) \leq \sqrt{\lambda_{\max}(\Sigma_0)}\|v\|$. Combining with $\|\text{grad } \Phi(X)\| \leq L_c\|X - X^*\|$ near X^* gives the bound. Part (ii): $(I - P_k)$ is an orthogonal projector,

so $\text{grad}_X \|(I - P_k)X\|$ is a unit vector. Parts (iii)–(iv) follow from direct gradient computation using super-critical growth estimates (Theorem 4.3) and score-function lower bounds. $\square$

3.2. Gradient Alignment. LEMMA 3.9 (Gradient of E_{BS}). *Under Assumption 3.3, E_{BS} is C^1 with*

$$\text{grad } E_{\text{BS}}(X) = \underbrace{\sum_{i=1}^4 \psi'_i(\delta_i) \text{grad}_X \delta_i}_{\text{grad } E_{\text{BS}}^{\text{dev}}} + \underbrace{\text{grad}_X \sum_{i < j} \phi(\|x_i - x_j\|)}_{\text{grad } E_{\text{BS}}^{\text{pair}} = -F_{\text{pair}}}.$$

LEMMA 3.10 (Upper bound). *Under Assumptions 3.1–3.3, on $\mathcal{L}_c$ there exists $c_2 > 0$ such that $\|\text{grad } E_{\text{BS}}(X)\| \leq c_2 \|\text{grad } \Phi(X)\|$.*

LEMMA 3.11 (Lower bound). *Under Assumptions 3.1–3.4, on $\mathcal{L}_c$ there exists $c_1 > 0$ such that $\|\text{grad } E_{\text{BS}}(X)\| \geq c_1 \|\text{grad } \Phi(X)\|$.*

Proofs of the upper and lower bounds follow from the triangle inequality, the Lipschitz properties of δ_i , and the separation condition. Full details are in Appendix A.

3.3. Theorem C: Full Statement and Proof. THEOREM 3.12 (Equivalence of Blind-Spot Geometry and Instability Energy). *Let Assumptions 3.1–3.5 hold and let X^* be a strict local minimum of Φ . Then on any sublevel set $\mathcal{L}_c$:*

(C1) **Morse-index equivalence.** *At every critical point $X_c \in \mathcal{L}_c$ (where $\text{grad } \Phi(X_c) = 0$ and $\text{grad } E_{\text{BS}}(X_c) = 0$),*

$$\text{ind}_{E_{\text{BS}}}(X_c) = \text{ind}_{\Phi}(X_c), \quad (3.1)$$

where $\text{ind}_f(X_c)$ denotes the Morse index (number of negative eigenvalues of $D^2 f(X_c)$).

(C2) **Topological energy equivalence.** *The sublevel sets $\mathcal{L}_c^{E_{\text{BS}}} := \{X : E_{\text{BS}}(X) \leq c\}$ and $\mathcal{L}_c^{\Phi} := \{X : \Phi(X) \leq c\}$ are homotopy equivalent for all regular values c . In particular, at every critical value c^* where the topology of the sublevel set changes,*

$$\Delta\chi(E_{\text{BS}})|_{c^*} = \Delta\chi(\Phi)|_{c^*}, \quad (3.2)$$

where $\Delta\chi$ denotes the jump in the Euler characteristic.

(C3) **Gradient norm equivalence.** *There exist $c_1, c_2 > 0$ with $c_1 \|\text{grad } \Phi\| \leq \|\text{grad } E_{\text{BS}}\| \leq c_2 \|\text{grad } \Phi\|$ and alignment $\cos \theta \geq \nu/c_2$.*

(C4) **Energy equivalence.** *There exist $a, b > 0$ with $aV(X) \leq E_{\text{BS}}(X) - E_{\text{BS}}(X^*) \leq bV(X)$, where $V(X) := \Phi(X) - \Phi(X^*)$.*

(C5) **Decoupled stabilisation.** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma(E_{\text{BS}}(X)) \text{grad } E_{\text{BS}}(X), \quad (3.3)$$

with $\gamma(E) \geq \kappa > 0$ for $E > \theta_0$ render X^ locally asymptotically stable. The Lyapunov v2 ISS certificate (Proposition 6.1) guarantees iteration convergence independently of the prediction signal: the Morse index $\text{ind}_{E_{\text{BS}}}(X^T) \geq 1$ serves as the instability alarm, decoupled from the descent certificate.*

Proof. Part (C1). By the gradient bounds (C3) and Assumption 3.5, X_c is a critical point of E_{BS} if and only if it is a critical point of Φ . At each such X_c , define the linear map $A := D^2 E_{\text{BS}}(X_c)$ and $B := D^2 \Phi(X_c)$. A second-order Taylor expansion of the gradient bound $c_1 \|\text{grad } \Phi\| \leq \|\text{grad } E_{\text{BS}}\| \leq c_2 \|\text{grad } \Phi\|$ near X_c yields $c_1^2 v^\top B^2 v \leq v^\top A^2 v \leq c_2^2 v^\top B^2 v$ for all $v \in \mathbb{R}^{Nd}$. Since Assumption 3.5

ensures both A and B are non-degenerate, the eigenvalues of A and B are jointly either positive or negative in each eigenspace. Sylvester's law of inertia then gives $\text{ind}_{E_{\text{BS}}}(X_c) = \text{ind}_{\Phi}(X_c)$.

Part (C2). By (C1) and Morse non-degeneracy, E_{BS} and Φ have identical critical points with identical Morse indices. Morse theory [20] establishes that the homotopy type of the sublevel set $\{f \leq c\}$ changes only at critical values, by attaching a cell of dimension equal to the Morse index. Since the critical values, indices, and attachment sequence are identical for E_{BS} and Φ , the sublevel sets are homotopy equivalent for all regular values c . The Euler characteristic jump at a critical value c^* with a single critical point of index k is $\Delta\chi = (-1)^k$; since k is the same for both functionals, (3.2) follows.

Part (C3). The bounds are Lemmas 3.10 and 3.11. For alignment: the proof of Lemma 3.11 (Appendix A) establishes $\langle \text{grad } E_{\text{BS}}, \text{grad } \Phi / \|\text{grad } \Phi\| \rangle \geq c_1 \|\text{grad } \Phi\| > 0$ by combining the shared pairwise term with monotonicity (S2). Dividing by $\|\text{grad } E_{\text{BS}}\| \leq c_2 \|\text{grad } \Phi\|$ gives $\cos \theta = \langle \text{grad } E_{\text{BS}}, \text{grad } \Phi \rangle / (\|\text{grad } E_{\text{BS}}\| \|\text{grad } \Phi\|) \geq c_1 / c_2 > 0$.

The empirical value $\cos \theta = -0.335$ on real data is measured *globally* (not near X^*) where the local assumptions of this theorem need not hold; the norm equivalence holds throughout $\mathcal{L}_c$.

Part (C4). Since Φ is coercive with $H^* > 0$ (Assumptions 3.1–3.2), the sublevel sets $\mathcal{L}_c$ are star-convex with respect to X^* for sufficiently small c : the ray $s \mapsto X^* + s(X - X^*)$, $s \in [0, 1]$, lies in $\mathcal{L}_c$ whenever $X \in \mathcal{L}_c$. Applying the fundamental theorem of calculus along this ray to both $E_{\text{BS}}(X) - E_{\text{BS}}(X^*)$ and $V(X) = \Phi(X) - \Phi(X^*)$ and using the gradient bounds (C3) pointwise along the ray, the two-sided energy bound follows with $a = c_1$, $b = c_2$. (For the lower bound, monotonicity (S2) is used to ensure the integrand remains non-negative along the ray.)

Part (C5). Take $W(X) = E_{\text{BS}}(X) - E_{\text{BS}}(X^*)$ as the Lyapunov candidate. By (C4), $aV \leq W \leq bV$, so W is positive definite and proper. Along (3.3):

$$\begin{aligned} \dot{W} &= \langle \text{grad } E_{\text{BS}}, F \rangle - \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2 \\ &\leq -\frac{\cos \theta}{c_2} \|\text{grad } E_{\text{BS}}\|^2 - \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2 \leq -\kappa \|\text{grad } E_{\text{BS}}\|^2 \leq 0. \end{aligned}$$

LaSalle's invariance principle [16] on the compact set $\{W \leq W(X_0)\}$ concludes convergence to X^* .

The *decoupling* is the key structural advance: the ISS certificate (Proposition 6.1) ensures that the Euler integration converges regardless of whether a crisis is occurring. The prediction signal is the Morse index of the converged configuration X^T : $\text{ind}_{E_{\text{BS}}}(X^T) = 0$ at normal equilibria (the minimum X^*) versus $\text{ind}_{E_{\text{BS}}}(X^T) \geq 1$ near saddle points (transitions out of the basin of attraction). This separation is topological and does not depend on the magnitude of the Lyapunov descent rate. $\square$

THEOREM 3.13 (Entropy Structure of BSDT). *Under Assumptions 3.1–3.4, the functional E_{BS} defines a Lyapunov entropy for the coupled dynamical system (2.1) in the following sense:*

(E1) **Monotone dissipation.** *Along every trajectory of (3.3),*

$$\frac{d}{dt} E_{\text{BS}}(X(t)) \leq 0.$$

(E2) **Non-negative entropy production.** *The entropy production rate*

$$\sigma(X) := -\frac{d}{dt} E_{\text{BS}}(X(t)) = \gamma(E_{\text{BS}}(X)) \|\text{grad } E_{\text{BS}}(X)\|^2 \geq 0.$$

(E3) **Equilibrium characterisation.** $\sigma(X) = 0$ if and only if $X = X^*$.

Consequently, the system (3.3) is a generalised Onsager gradient flow with positive-semidefinite mobility $M(X) = \gamma(E_{\text{BS}}(X)) I \succeq 0$.

Proof. (E1) follows directly from Part (C5) of Theorem 3.12: $\dot{W} \leq -\kappa \|\text{grad } E_{\text{BS}}\|^2 \leq 0$.

(E2) Differentiating $W = E_{\text{BS}} - E_{\text{BS}}(X^*)$ along (3.3):

$$\dot{W} = \langle \text{grad } E_{\text{BS}}, F - \gamma(E_{\text{BS}}) \text{grad } E_{\text{BS}} \rangle = \langle \text{grad } E_{\text{BS}}, F \rangle - \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2.$$

By (E1), $\dot{W} \leq 0$, so rearranging: $\sigma(X) := -\dot{W} = \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2 - \langle \text{grad } E_{\text{BS}}, F \rangle \geq 0$. (Note: $\langle \text{grad } E_{\text{BS}}, F \rangle$ can be positive in general—the empirical anti-alignment $\cos \theta = -0.335$ reflects this—but the damping term $\gamma \|\text{grad } E_{\text{BS}}\|^2$ dominates; the non-negativity of σ is a consequence of (C5), not an independent assumption.)

(E3) $\sigma(X) = 0$ requires $\dot{W} = 0$. From (C5), $\dot{W} = \langle \text{grad } E_{\text{BS}}, F \rangle - \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2 = 0$. At a non-equilibrium point $X \neq X^*$, $\|\text{grad } E_{\text{BS}}\| > 0$ by Lemma 3.11 and $\gamma(E_{\text{BS}}) > 0$ (since $E_{\text{BS}}(X) > E_{\text{BS}}(X^*)$ by (C4) and $\gamma(E) > 0$ for $E > E_{\text{BS}}(X^*)$ by the strict positivity of $E/(E+\theta)$). Hence $\sigma(X) > 0$ for all $X \neq X^*$, and $\sigma(X^*) = 0$ since $\text{grad } E_{\text{BS}}(X^*) = 0$ by (C3). The Onsager structure is immediate from the form of (3.3). $\square$

REMARK 3.14. *The entropy production $\sigma(X)$ provides an operational definition of the distance to equilibrium that is computable in real time from market data. For the fluid extension (Section 9), σ corresponds to the production/dissipation gap $P - D$ in (9.5); the P/D halving principle (Table 9.2) is the numerical signature of $\sigma \rightarrow 0$ as the adaptive viscosity drives the system toward the entropy-neutral manifold $P = D$.*

REMARK 3.15. *The lower bound $c_1 \|\text{grad } \Phi\| \leq \|\text{grad } E_{\text{BS}}\|$ is essential for (C5): without it, $\text{grad } E_{\text{BS}}$ could be orthogonal to F , and the damping term would exert no stabilising force. The separation condition (Assumption 3.4) is the minimal non-degeneracy requirement for detection-based stabilisation. The Morse-index equivalence (C1) provides the topological counterpart: the prediction signal (index ≥ 1) is structurally invariant under small perturbations, unlike the gradient magnitude which depends on the constants c_1, c_2 .*

4. Adaptive Damping and the Critical Manifold. THEOREM 4.1 (Monotone Descent). *Under Assumption 3.1, $\frac{d}{dt} \Phi(X(t)) = -\|\text{grad } \Phi\|^2 \leq 0$. Every ω -limit point satisfies $\text{grad } \Phi = 0$.*

THEOREM 4.2 (Local Exponential Stability). *Under Assumptions 3.1–3.2, X^* is locally exponentially stable with rate $\lambda_{\min}(H^*)$:*

$$\|X(t) - X^*\| \leq \|X(0) - X^*\| e^{-(\alpha - L_{\text{pair}})t} \quad \text{for } \alpha > L_{\text{pair}}.$$

THEOREM 4.3 (Spectral Phase Transition). *Define the critical manifold*

$$C^* := \{X : \lambda_{\max}(\rho(X) \ell(X) \mathbf{W}) = 1\}, \quad (4.1)$$

where $\rho(X)$ is mean pairwise correlation, $\ell(X)$ mean exposure, and $\mathbf{W}$ the row-normalised adjacency matrix. Equivalently, C^* is the locus where the Morse index of the equilibrium jumps from 0 (stable minimum) to ≥ 1 (saddle or unstable):

$$C^* = \{X : \text{ind}_{\Phi}(X^*(X)) \text{ changes from } 0 \text{ to } \geq 1\}, \quad (4.2)$$

where $X^*(X)$ is the nearest equilibrium to X .

- (i) *Below $\mathcal{C}^*$: $\text{ind}_\Phi(X^*) = 0$, and for any fixed coupling $\bar{\gamma}$, the gradient flow is locally asymptotically stable.*
- (ii) *Above $\mathcal{C}^*$: $\text{ind}_\Phi(X^*) \geq 1$, and for any fixed $\bar{\gamma} > 0$, there exists a configuration X^+ with a growing mode. No constant $\bar{\gamma}$ stabilises all above- $\mathcal{C}^*$ configurations.*
- (iii) *Adaptive $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ achieves marginal stability everywhere and keeps $\text{ind}_\Phi(X^*) \leq 0$ outside a neighbourhood of $\mathcal{C}^*$.*

Proof. Part (i). Below $\mathcal{C}^*$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\text{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [18] gives LAS.

Part (ii). Above $\mathcal{C}^*$, a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ has eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^*, u \rangle^2$ satisfies $\dot{W} \geq (\lambda_u - \alpha)W + O(\|X - X^*\|^3)$. For initial data $\|X(0) - X^*\| < r_0$ with r_0 sufficiently small (so that the cubic remainder is dominated by the linear term), forward Gronwall gives $W(t) \geq W(0)e^{(\lambda_u - \alpha)t/2}$, growing exponentially.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ zeroes the net spectral abscissa by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}^*$ follows from the radial spring dominating once $\lambda_{\max} < \alpha$. $\square$

COROLLARY 4.4 (Basel III Proccyclicality). *A fixed regulatory capital buffer $\bar{\kappa}$ corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 4.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}^*$), no fixed $\bar{\kappa}$ simultaneously maintains conservatism in normal times and sufficiency in stress.*

5. Online Algorithm and Optimal Policy.

5.1. GravityEngine as Online Convex Optimisation. At each step t , the regulator selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$; the system evolves to $X_{t+1}(\gamma_t)$ and cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed.

PROPOSITION 5.1 (Online Regret Bound). *If c_t is convex and L_c -Lipschitz in γ , projected online gradient descent achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^*(X^t)) \leq L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}. \quad (5.1)$$

Proof. $X^{t+1}(\gamma)$ is affine in γ ; composing the convex Φ yields convexity. The Lipschitz bound $|\partial_\gamma c_t| \leq \eta F_{\max}^2 =: L_c$ follows from the force clamp. Standard OGD guarantees [12, 23] then apply. $\square$

PROPOSITION 5.2 (Computational Complexity). *Computing $\text{grad } E_{\text{BS}}(X)$ costs $O(N^2d)$ per step. With k -d tree acceleration and graph sparsification, complexity reduces to $O(Nd \log N)$.*

PROPOSITION 5.3 (Reduced Tensor Descriptor). *Let $\mathcal{D}(X) \in \mathbb{R}^{d+7}$ be the reduced descriptor:*

$$\mathcal{D}(X) = (\|\text{grad } E_{\text{BS}}\|, \lambda_1(H), \dots, \lambda_d(H), \delta_C(X), \delta_{\text{med}}(X), \text{tr}(H), \det(\Sigma)), \quad (5.2)$$

where $H = D^2E_{\text{BS}}(X)$, δ_{med} is the medoid distance, and Σ is the local covariance. Then:

- (i) $\mathcal{D}$ is computable in $O(Nd + d^3)$ time (gradient: $O(Nd)$; Hessian eigenvalues via Lanczos on the $d \times d$ marginal: $O(d^3)$; statistics: $O(Nd)$).
- (ii) $\mathcal{D}$ is locally injective near X^* and near $\mathcal{C}^*$: distinct states $X \neq X'$ in a neighbourhood of X^* (resp. $\mathcal{C}^*$) satisfy $\mathcal{D}(X) \neq \mathcal{D}(X')$.
- (iii) The Morse-index alarm $\text{ind}_{E_{\text{BS}}} \geq 1$ is fully determined by $\mathcal{D}$ via the sign pattern of $(\lambda_1, \dots, \lambda_d)$.

Proof. (i): The gradient $\text{grad } E_{\text{BS}}$ is a sum of N terms, each $O(d)$; the Hessian eigenvalues are computed on the d -dimensional marginal (averaging over agents) via Lanczos iteration. (ii): Near X^* , the tuple $(\|\text{grad } E_{\text{BS}}\|, \delta_C, \delta_{\text{med}})$ separates radial distance; the Hessian eigenvalues separate angular position. Near $\mathcal{C}^*$, the leading eigenvalue λ_1 crosses zero, providing a discriminating coordinate. (iii): $\text{ind}_{E_{\text{BS}}} = \#\{j : \lambda_j < 0\}$ by definition. $\square$

Algorithm 1 Lyapunov v2 Barrier-Augmented Euler Integration with Decoupled Morse Alarm

Require: State X^0 , energy V , force F , step η_0 , Armijo $c = 10^{-4}$, backtrack $\rho = 0.5$, La Salle tolerance $\tau_{\text{LS}} = 10^{-5}$, patience $P = 5$

```

1:  $p \leftarrow 0$  {La Salle counter}
2: {Phase 1: ISS-certified iteration (convergence)}
3: for  $t = 0, 1, \dots, T - 1$  do
4:    $F^t \leftarrow F(X^t)$ ;  $F^t \leftarrow M \tanh(F^t/M)$   $\{C^1$  barrier clamp,  $M = 10\}$ 
5:    $V_{\text{old}} \leftarrow V(X^t)$ ;  $g \leftarrow \|F^t\|^2$ 
6:    $X^{\text{cand}} \leftarrow X^t + \eta_t F^t$ 
7:   Compute ISS perturbation  $\|w^t\|$  (pairwise force norm)
8:   while  $V_{\text{old}} - V(X^{\text{cand}}) < c \eta_t g - \|w^t\|^2$  do
9:      $\eta_t \leftarrow \rho \eta_t$ ;  $X^{\text{cand}} \leftarrow X^t + \eta_t F^t$  {backtrack}
10:  end while
11:   $X^{t+1} \leftarrow X^{\text{cand}}$ ;  $\eta_{t+1} \leftarrow \min(\eta_t/\rho, \eta_0)$  {optimistic step-size recovery}
12:  if  $|\Delta V|/(|V| + 10^{-10}) < \tau_{\text{LS}}$  then
13:     $p \leftarrow p + 1$ ; if  $p \geq P$  break {La Salle}
14:  else
15:     $p \leftarrow 0$ 
16:  end if
17: end for
18: {Phase 2: Morse-index prediction (alarm)}
19: Compute Hessian eigenvalues  $\lambda_1 \leq \dots \leq \lambda_{Nd}$  of  $D^2 E_{\text{BS}}(X^T)$ 
20:  $\text{ind} \leftarrow \#\{j : \lambda_j < 0\}$  {Morse index}
21: if  $\text{ind} \geq 1$  then
22:   Flag instability alarm {saddle  $\Rightarrow$  phase transition}
23: end if
24: Score  $X^T$  via FusedSystemScorer ((6.6), (6.7))
Ensure:  $V(X^T) \leq V(X^0)$  {ISS descent guaranteed}

```

5.2. Optimal Policy via Dynamic Programming. The social planner minimises total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t (E(X_t) + \kappa \gamma_t^2) \mid X_0 = X \right], \quad (5.3)$$

where $\kappa > 0$ is the cost of intervention and $\beta \in (0, 1)$ the discount factor.

THEOREM 5.4 (Optimal Adaptive Policy). *The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \approx \frac{\beta \|\text{grad } E(X)\|^2}{2\kappa + \beta \|\text{grad } E(X)\|^2} \cdot \frac{E(X)}{E(X) + \theta}, \quad \theta = \kappa/\beta. \quad (5.4)$$

This recovers the adaptive damping law $\gamma(E) = E/(E + \theta)$ as its leading-order term.

THEOREM 5.5 (Necessity of State-Dependence). *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}^*$:*

$$V^*(X) - V_{\bar{\gamma}}(X) \geq \Delta(\rho(X), \ell(X)) > 0.$$

Constant friction is strictly suboptimal above $\mathcal{C}^*$.

6. System Mode Engines. The gradient flow (2.1) admits multiple physically-motivated pairwise potentials. We introduce two complementary engines—*Molecular* and *Gravity*—that share a common integration controller and post-simulation scoring stack, differing only in the force law.

6.1. Molecular Engine (Lennard-Jones 6–12). Replace the Gaussian/log potential in (2.2) with the Lennard-Jones 6–12 pair potential:

$$\phi_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad (6.1)$$

with well depth $\varepsilon = 1$, particle diameter $\sigma = 1$, radial centering spring $\alpha_{\text{rad}} = 0.1$, step size $\eta = 0.01$, and $T = 80$ iterations. The pairwise force on particle i from neighbour j is

$$F_{ij} = \frac{24\varepsilon}{r_{ij}} \left[2 \left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] \hat{r}_{ij}, \quad (6.2)$$

truncated to $k = 15$ nearest neighbours for $O(nk)$ per-step cost. Normal points settle into minimum-energy crystalline configurations; anomalies are expelled to high-energy positions.

6.2. Gravity Mode Engine. The Gravity Mode Engine retains the Gaussian attraction with Coulomb-type repulsion:

$$F_{ij}^{\text{grav}} = -\gamma \left[\exp\left(-\frac{r_{ij}^2}{\sigma^2}\right) - \frac{\lambda_{\text{rep}}}{r_{ij}} \right] \hat{r}_{ij}, \quad (6.3)$$

with $\alpha = 0.1$, $\gamma = 0.5$, $\sigma = 1$, $\lambda_{\text{rep}} = 0.05$, $\eta = 0.05$, $T = 60$. The Lyapunov energy candidate uses radial energy only: $V(X) = \frac{\alpha}{2} \sum_i \|x_i - \mu\|^2$, with the pairwise force treated as a bounded perturbation under the ISS framework (Proposition 6.1).

6.3. Hybrid Engine. The Hybrid Gravity Engine runs both engines independently and returns the convex combination

$$s_{\text{hyb}} = w^* s_{\text{mol}} + (1 - w^*) s_{\text{grav}}, \quad (6.4)$$

where $w^* \in \{0, 0.1, \dots, 1.0\}$ is selected by grid search maximising cross-validated AUROC.

6.4. Lyapunov v2 Stability Controller. All three engines share a common *Lyapunov v2 stabiliser* that controls the Euler integration. The controller provides a three-part convergence certificate.

PROPOSITION 6.1 (Lyapunov v2 Three-Part Convergence). *Let $V : \mathbb{R}^{Nd} \rightarrow \mathbb{R}$ be the Lyapunov energy candidate (LJ energy for the Molecular Engine, radial energy for the Gravity Engine). Then the barrier-augmented Euler integration with step size η_t satisfies:*

- (i) **Armijo–ISS descent.** $V(X^t) - V(X^{t+1}) \geq c\eta_t \|\text{grad } V\|^2 - \|w^t\|^2$, where $c = 10^{-4}$ is the Armijo constant and $\|w^t\|$ bounds the exogenous perturbation (pairwise force for Gravity, zero for Molecular).
- (ii) **La Salle invariance.** Iteration terminates when the energy-change ratio $|\Delta V|/(|V| + 10^{-10}) < \tau_{\text{LS}} = 10^{-5}$ persists for five consecutive steps.
- (iii) **Coercivity.** V is bounded below and coercive: $V(X) \rightarrow +\infty$ as $\|X\| \rightarrow \infty$.

Consequently, the sequence $\{V(X^t)\}$ converges monotonically and every limit point satisfies $\|\text{grad } V\| \leq \|w^\infty\|$.

Proof. Part (i): Write the total force as $F^t = -\text{grad } V(X^t) + w^t$, where $-\text{grad } V$ is the conservative (radial + LJ gradient) force and w^t is the residual perturbation (for the Gravity Engine, w^t is the pairwise force not captured by the radial Lyapunov function; for the Molecular Engine, $w^t = 0$ since V equals the full LJ energy and all kNN-limited forces are consistent with $\text{grad } V$). Taylor expansion gives

$$V(X^{t+1}) = V(X^t) + \eta_t \langle \text{grad } V, F^t \rangle + O(\eta_t^2 \|F^t\|^2).$$

Substituting $F^t = -\text{grad } V + w^t$:

$$V(X^t) - V(X^{t+1}) = \eta_t \|\text{grad } V\|^2 - \eta_t \langle \text{grad } V, w^t \rangle - O(\eta_t^2 \|F^t\|^2).$$

The Armijo backtracking reduces η_t until the $O(\eta_t^2)$ term is small: specifically, η_t is halved until $V(X^t) - V(X^{\text{cand}}) \geq c\eta_t \|F^t\|^2 - \|w^t\|^2$. By Cauchy–Schwarz, $|\langle \text{grad } V, w^t \rangle| \leq \|\text{grad } V\| \|w^t\|$, so the cross-term is absorbed into $\|w^t\|^2$ with a constant adjustment. The resulting Armijo–ISS condition is: $V(X^t) - V(X^{t+1}) \geq c\eta_t \|\text{grad } V\|^2 - \|w^t\|^2$.

Part (ii): The La Salle patience counter certifies that the sequence has entered the invariant set $\{X : \|\text{grad } V(X)\| \leq \tau_{\text{LS}}\}$. Part (iii): The radial spring $\alpha\|x_i - \mu\|^2/2$ and the r^{-12} repulsive core (Molecular) or $\log(r)$ repulsion (Gravity) jointly ensure $V(X) \rightarrow +\infty$. $\square$

REMARK 6.2. The $O(nk)$ per-step cost (k NN-limited interactions with $k = 15$) ensures the stabiliser is practical for $n \leq 3000$ points. The backtracking parameter $\rho \approx 0.65$ in practice; the barrier clamp uses a C^1 -smooth function $f(x) = M \tanh(x/M)$ with $M = 10$ to bound forces without introducing discontinuities.

6.5. Fused Topological Scoring Architecture. Both engines share an identical post-simulation scoring pipeline. After physics simulation separates anomalies structurally, three complementary signal families are extracted and fused.

Morse Critical-Point Alarm.. The MorseTopologyAlarm computes four features per point: mean k NN distance, local intrinsic dimension d_1 , persistence proxy, and local density ratio, weighted $w = (0.35, 0.30, 0.20, 0.15)$:

$$s_{\text{Morse},i} = \sum_{j=1}^4 w_j \max(z_{ij}, 0), \quad (6.5)$$

where z_{ij} is the z-score of feature j relative to the normal reference set.

Betti Barcode Suite.. The BettiBarcodeSuite ($k = 20$, 8 scales) extracts 19 topological features: β_0 (1-reach: disconnected fraction), β_1 (reach $\times$ CV: excess connectivity), Euler characteristic $\chi = \beta_0 - \beta_1$, and Conley stability (CoV of β_0 across scales).

UDL Post-Simulation Operators.. Four mathematical operators—PhaseCurve (sequential trajectory structure), Topological (LID, persistence), KernelRKHS (non-linear manifold deviation), and RankOrder (distribution-free extremity)—produce a ~ 27 -dimensional feature vector. The anomaly score is the RMS z-deviation:

$$s_{\text{UDL},i} = \sqrt{\frac{1}{D} \sum_{d=1}^D z_{id}^2}.$$

Score Fusion.. The *FusedSystemScorer* combines these three signal families by equal-weight averaging after min-max normalisation:

$$s_{\text{fused}} = \frac{1}{3} (\bar{s}_{\text{Morse}} + \bar{s}_{\text{Betti}} + \bar{s}_{\text{UDL}}), \quad (6.6)$$

where $\bar{s}$ denotes min-max normalisation to $[0, 1]$. For large sample sizes ($n \geq 500$), an enriched weighting $0.4 \times s_{\text{Morse}} + 0.6 \times s_{\text{enriched}}$ is used.

SpectraFalseAlarmFilter.. A persistence-gated sigmoid filter suppresses false alarms on points whose local topology is insufficiently persistent:

$$g_i = \text{sigmoid}(10 \cdot (p_i - \theta_p)), \quad s_{\text{filtered},i} = g_i \cdot s_{\text{fused},i}, \quad (6.7)$$

where p_i is the local persistence and $\theta_p = Q_{0.95}(\{p_j\}_{j \in \text{ref}})$.

FAR-Targeted Calibration.. The *FARTargetCalibrator* enforces a user-specified false-alarm rate (default FAR = 0.05) via two-stage piecewise-linear rescaling:

$$s_{\text{out}} = \begin{cases} \frac{1}{2} s / \hat{\tau} & s \leq \hat{\tau}, \\ \frac{1}{2} + \frac{1}{2} (s - \hat{\tau}) / (1 - \hat{\tau}) & s > \hat{\tau}, \end{cases} \quad (6.8)$$

where $\hat{\tau}$ is the empirical quantile threshold corresponding to the target FAR.

6.6. Universal Conformal Scoring. The fused topological score (6.6) is a dimensionless ranking statistic whose distribution depends on the specific engine and dataset. To obtain *distribution-free* statistical guarantees, we add a conformal prediction layer that converts raw scores into valid p -values under the sole assumption of exchangeability.

Split conformal calibration.. Given a labelled reference set $(X_{\text{ref}}, y_{\text{ref}})$, we split into training $(1 - \alpha_{\text{cal}})$ and calibration $(\alpha_{\text{cal}} = 0.25)$ folds. The engine is fitted on the training fold; the resulting scores on the calibration fold form the *nonconformity distribution* $\{R_j\}_{j=1}^{n_{\text{cal}}}$. For a new observation $\mathbf{x}$, the conformal p -value is

$$p(\mathbf{x}) = \frac{|\{j : R_j \geq s(\mathbf{x})\}| + 1}{n_{\text{cal}} + 1}, \quad (6.9)$$

where $s(\mathbf{x})$ is the engine's raw anomaly score. By the exchangeability lemma [34], $\Pr(p \leq \alpha) \leq \alpha$ for any significance level $\alpha \in (0, 1)$.

Panel aggregation.. For time-structured data partitioned into Q panels (e.g. quarters), we compute six aggregation features per panel: mean, 90th percentile, max, standard deviation, rolling z -score momentum, and score momentum. These are rank-normalised and max-fused to produce a panel-level anomaly score. The final *universal score* is

$$s_{\text{univ}} = \max(\tilde{s}_{\text{point}}, \tilde{s}_{\text{panel}}), \quad (6.10)$$

where $\tilde{s}$ denotes rank-normalisation to $[0, 1]$. This meta-strategy automatically selects the better view: point-level scoring dominates on high-frequency data (ERCOT), while panel-level aggregation dominates on structural panel data (G-SIB quarterly balance sheets).

Key property: conformal scoring never degrades.. Since the universal score takes the *max* of the rank-normalised point and panel views, it is guaranteed to be at least as discriminative as the better of the two: $\text{AUC}(s_{\text{univ}}) \geq \max(\text{AUC}(s_{\text{point}}), \text{AUC}(s_{\text{panel}}))$. This holds because rank-normalisation is a monotone transformation (preserving AUC) and the pointwise max of two scores preserves any separation that either individual score achieves.

7. Empirical Validation.

7.1. Data. We construct $X(t) \in \mathbb{R}^{N \times d}$ from FDIC Statistics on Depository Institutions (SDI) quarterly call reports. $N = 7$ FDIC institution types, $d = 6$ features: loan-to-asset (LNLSNET/ASSET), equity ratio (EQ/ASSET), NPL ratio (NCLNLS/LNLSNET), ROA (NETINC/ASSET), funding cost (EINTEXP/LNLSNET), and yield-curve slope (FRED T10Y2Y). The panel spans $T = 140$ quarters (1990-Q1 to 2024-Q4). The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters). The inter-sector exposure network $\mathbf{W}$ is estimated by Ledoit–Wolf Oracle shrinkage [17] ($\rho^* = 0.046$; spectral radius $\lambda_{\max}(\mathbf{W}) = 2.98$).

7.2. Gradient Alignment on Real Data. The per-quarter cosine alignment $\cos \theta(t) = \langle \text{grad } E_{\text{BS}}, -\text{grad } \Phi \rangle / (\|\text{grad } E_{\text{BS}}\| \|\text{grad } \Phi\|)$ is *negative* (mean = -0.335) across all 140 quarters. This anti-alignment is significant: the deviation gradient and the restoring force measure *complementary* axes of instability. The potential Φ pulls institutions toward their current cluster centre (spatial cohesion); E_{BS} measures deviation from the *historical* normal (temporal anomaly). The equivalence theorem holds in the mathematical sense (both are Lyapunov candidates), but their gradients span complementary subspaces on real data.

7.3. Critical Manifold and Super-Criticality. The system spends **68.6%** of the 140-quarter sample above $\mathcal{C}^*$, implying that the U.S. banking system is *structurally over-coupled*. Crises occur when perturbations arrive while the system is already super-critical, consistent with the persistence view of systemic risk [8].

TABLE 7.1
Lead time of MFLS versus benchmark stress indices (quarters, positive = MFLS precedes benchmark). Data: FDIC SDI, $T = 140$.

Crisis Window	Benchmark	Lead (quarters)
GFC 2008	STLFSI	+3
	VIX	+1
	NFCI	+2
COVID 2020	STLFSI	+3
	VIX	+3
Rate Shock 2022	STLFSI	0
	VIX	+1

7.4. Crisis Window Analysis. The MFLS signal leads STLFSI by 3 quarters during the GFC, firing in 2007-Q1—six quarters before the Lehman collapse.

7.5. Out-of-Sample Backtest. A single P75 threshold trained on 1990-Q1 to 2005-Q4 (no crisis data) is evaluated on 2006-Q1 to 2024-Q4:

TABLE 7.2

Out-of-sample backtest. Single P75 threshold trained on pre-2006 data.

Crisis	First alarm	Lead	Hit rate	FAR
GFC	2007-Q1	6Q	71.4%	35%
COVID	2019-Q4	1–2Q	62.5%	35%
Rate Shock	2022-Q1	coincident	50.0%	33%
AUROC ≥ 0.70 over full 2006–2024 test window.				

7.6. Universal Granger Failure. All five MFLS scoring variants—Baseline (Mahalanobis), Full-BSDT, QuadSurf, Signed-Fisher, Expo-Gate—fail Granger causality at $\alpha = 0.10$ ($p > 0.21$ at all lags). This is not an artefact but the *expected signature* of a phase-transition detector: if crises are abrupt threshold crossings rather than trends, Granger must fail. The MFLS signal is persistently elevated for years without a crisis (the system sits near the stability boundary), then one perturbation triggers the transition. The relationship is *nonlinear in time*.

7.7. Institution-Level Extension. Replication on the top-30 U.S. banks by total assets (CERT-level data) produces stronger results: AUROC rises to 0.805, GFC hit rate to 75.0%, and $\lambda_{\max}(\mathbf{W}) = 12.1$ (vs. 8.3 for SPECGRP), reflecting denser interconnection among individual banks. The Granger failure is universal at the institution level.

7.8. Sensitive-Sector Real-World Validation. To test the System Mode engines on high-stakes domains, we apply the Hybrid Engine (Section 6.3) with FARTargetCalibrator to three sensitive-sector datasets.

FRED Macroeconomic Indicators.. $d = 6$ recession-sensitive features (T10Y2Y yield-curve slope, UNRATE, ICSA initial claims, UMCSSENT consumer sentiment, INDPRO industrial production, PERMIT building permits) at monthly frequency, 1990–2024. NBER recession quarters serve as ground truth. The Hybrid Engine achieves AUC = 0.9999.

ERCOT Energy-Grid Dispatch.. Hourly generation-by-fuel data for the Texas ERCOT interconnect, $d = 5$ features (wind, solar, gas, coal, nuclear capacity factors). Anomaly labels from the February 2021 Winter Storm Uri blackout window. The Hybrid Engine achieves AUC = 0.9940.

Prospective G-SIB Bank-Level Panel.. Quarterly balance-sheet data for 30 Globally Systemically Important Banks (G-SIBs), $d = 6$ features (loan-to-asset, equity ratio, NPL ratio, ROA, funding cost, yield-curve slope). Using a strictly prospective protocol—reference calibrated on 1994:Q1–2003:Q4, tested on 2006:Q1–2024:Q4—the MFLS signal exceeds $z > 2$ at 2007:Q1, six quarters before the Lehman collapse. Per-bank ranking identifies Bear Stearns, Washington Mutual, and Countrywide in the top decile; COVID-2020 is correctly absent from the alarm set.

7.9. Prospective G-SIB Engine Comparison. To compare the three System Mode engines on a common benchmark, we run a prospective expanding-window evaluation on 25 G-SIBs over 76 quarters (2005:Q1–2023:Q4). All engines use $y = 0$ (no crisis labels) during training and scoring; label information enters only for ex-post evaluation metrics. The engines use identical calibration (reference period 1994:Q1–2003:Q4) and the FARTargetCalibrator with target FAR = 5%, calibrated on the pre-crisis reference window only. At every quarter t , scoring and threshold updates

TABLE 7.3
Sensitive-sector validation with System Mode engines.

Dataset	n	d	AUC	Engine
FRED Macro	1400	6	0.9999	Hybrid
ERCOT Grid	8760	5	0.9940	Hybrid
G-SIB Panel	4200	6	AUROC ≥ 0.80	Hybrid

are computed from data available up to t only; no forward-looking features, future labels, or hindsight parameter re-fitting are permitted.

TABLE 7.4
Prospective G-SIB engine comparison ($n = 25$ banks, $T = 76$ quarters, 2005–2023). Strictly causal protocol: no future data is used; crisis labels are used only for ex-post metric evaluation. FAR computed on non-crisis quarters only.

Engine	FAR	z -score	GFC Lead	COVID Alarm
Gravity	1.6%	+4.65	5Q	None
Molecular	14.3%	+2.23	4Q	None
Hybrid	7.9%	+2.65	5Q	None

The Gravity Engine dominates: FAR = 1.6% (vs. 14.3% Molecular) at $z = +4.65$, with 5-quarter GFC lead and zero COVID false alarm. The earliest per-bank flag is BNP Paribas at 2007:Q2, consistent with the August 2007 BNP Paribas fund freeze that marked the onset of the European dimension of the crisis. Chinese G-SIBs (ICBC, CCB, BOC, ABC) show the lowest mean scores (0.046), reflecting their state-capitalised, domestically focused balance sheets during this period.

7.10. Conformal Engine Benchmark. To evaluate the universal conformal scoring layer (Section 6.6), we benchmark all three engines in both standard (s_{fused}) and universal (s_{univ}) modes on two complementary datasets: ERCOT energy-grid dispatch (high-frequency, hourly, $n \approx 15,600$) and the G-SIB bank-level panel (low-frequency, quarterly, $n \approx 1,900$).

TABLE 7.5
Standard vs. universal conformal scoring: AUC-ROC across two sensitive-sector benchmarks. “Univ” denotes the max-fused point/panel score with conformal p -values (6.10). The “Lift” column reports the AUC improvement from conformal panel aggregation (negative = panel view was weaker but max-fusion preserved AUC).

Dataset	Engine	Standard	Universal	Lift (pp)
ERCOT	Gravity	0.9948	0.9948	0.0
	Molecular	0.9939	0.9939	0.0
	Hybrid	0.9997	0.9997	0.0
G-SIB	Gravity	0.766	0.806	+4.0
	Molecular	0.839	0.839	0.0
	Hybrid	0.814	0.818	+0.4

Three findings emerge. **(i)** On ERCOT, where anomalies are concentrated in a single temporal block (Winter Storm Uri), point-level scores already achieve near-

perfect separation; the universal layer preserves AUC exactly via the max-fusion guarantee. **(ii)** On the G-SIB bank-level panel, quarter-level aggregation lifts the Gravity Engine from 0.766 to 0.806 (+4.0 pp): the panel view detects slow-building financial stress that individual quarterly snapshots miss. The Hybrid Engine also improves (0.814 $\rightarrow$ 0.818), confirming that panel aggregation provides complementary signal. **(iii)** No engine regresses under universal scoring, validating the monotonicity property of Eq. (6.10).

These results confirm that conformal scoring is a *free* upgrade: it adds distribution-free p -values and panel-level aggregation at negligible computational cost, with guaranteed non-degradation and meaningful improvement on structural panel data.

7.11. Welfare Calibration. Translating $\gamma^*(X_t)$ into the Basel III CCyB format:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1} \gamma^*(X_t) \sigma_\ell(t). \quad (7.1)$$

The GFC window produces a consumption-equivalent welfare loss of 19.57 pp with peak CCyB of 152 bps. COVID shows near-zero welfare loss (crisis averted by emergency policy) despite peak CCyB of 231 bps. The 2022 rate shock produces $\mathcal{L} = 1.22$ pp with CCyB = 0 (absorbed without systemic transmission).

7.12. In-Sample Monitoring with Adaptive Calibration. A common methodological concern with systemic-risk benchmarks is that future information may leak into model training—through shared normalisation statistics, threshold selection on test data, or hyperparameter tuning on crisis periods. We address this with a comprehensive evaluation of eight algorithm variants under a *strictly prospective, in-sample monitoring* protocol with adaptive calibration.

Protocol. N-body physics engines (Molecular, Gravity, Hybrid) embed the *current* quarter into the particle simulation alongside all historical quarters. No future data is ever used: the engines receive $y = \mathbf{0}$ (no crisis labels) and all scoring, BSDT weighting, and threshold decisions depend solely on past and present observations. The detection signal arises from the interaction between the new observation and the historical configuration. The evaluation follows five rules:

1. **Expanding-window training.** For each quarter $t \in \{5, \dots, T-1\}$, the engine is trained on quarters $1, \dots, t$ including the current quarter. No future data is ever present in the training set.
2. **Preprocessing fit on past only.** A `StandardScaler` is fit on the training window $[1, t]$ and applied (transform only). No global statistics are used.
3. **Frozen hyperparameters.** All engine parameters (iterations = 60, k -neighbors = 10) are fixed from a pre-crisis validation period (2005 Q1–2007 Q3, $N_{\text{calib}} = 11$ quarters) and never updated.
4. **Adaptive threshold.** Because fixed thresholds suffer from score drift over 19-year horizons, we implement an *adaptive rolling percentile* strategy: the alarm threshold at time t is the 99th percentile of scores in a trailing 8-quarter window, $\tau_t = P_{99}(s_{t-8:t-1})$. This adapts to the non-stationary score distribution while using only past data.
5. **No label access at test time.** Crisis labels are used only for *ex-post* evaluation; the alarm decision at each quarter depends solely on the score and the adaptive threshold.

Algorithms evaluated. We benchmark eight algorithm variants spanning the full framework: (1) Molecular (base engine), (2) Gravity, (3) Hybrid (Molecular+Gravity blend), and five post-hoc variants combining the Molecular base with Fisher BSDT,

QuadSurf manifold, or ExpoGate exponential gating layers, as well as RTD (Reduced Tensor Descriptor) fusion with Molecular and Fisher layers.

Results.. Table 7.6 presents the comprehensive comparison across all eight algorithms with the Adaptive P99 threshold. The headline result is **Mol+ExpoGate**: AUROC = 0.867, with **zero false alarms** (FAR = 0.0%), recall = 38.5%, and precision = 100%. The first GFC alarm fires at 2007-Q4—two quarters *before* the Bear Stearns collapse and three quarters before Lehman Brothers. This represents a dramatic improvement over fixed-threshold approaches, where Mol+ExpoGate suffers FAR = 100% (every quarter triggers an alarm) due to score drift.

RTD+Fisher achieves the highest GFC-window AUROC (= 1.000) with recall = 23.1% at only 1.9% FAR, confirming that the tensor-descriptor fusion captures GFC dynamics with near-perfect discrimination. RTD+Mol provides early warning (first alarm 2007-Q4) with GFC-AUROC = 0.948.

Gravity alone (AUROC = 0.331) and pure Hybrid (AUROC = 0.607) cannot be rescued by any calibration strategy, confirming that the post-hoc layers and multi-view fusion are essential for competitive detection.

TABLE 7.6

*In-sample monitoring with Adaptive P99 threshold. T=76 quarters, N=25 G-SIBs, d=5 features, 13 crisis quarters. GFC 1st: quarter of first GFC alarm. Best values in each column are **bold**.*

Algorithm	AUROC	GFC-AUC	FAR%	Recall%	Prec.%	GFC 1st
Mol+ExpoGate	0.867	0.909	0.0	38.5	100.0	2007-Q4
RTD+Fisher	0.773	1.000	1.9	23.1	75.0	2007-Q4
Mol+QuadSurf	0.752	0.584	0.0	15.4	100.0	2008-Q3
RTD+Mol	0.730	0.948	1.9	15.4	66.7	2007-Q4
Mol+Fisher	0.725	0.701	3.8	15.4	50.0	2008-Q1
Molecular	0.722	0.753	9.6	23.1	37.5	2008-Q3
Hybrid	0.607	0.779	7.7	0.0	0.0	—
Gravity	0.331	0.299	5.8	0.0	0.0	—

The calibration problem and its solution.. Figure 7.1 illustrates the core calibration challenge: a fixed $\mu+3\sigma$ threshold computed in 2005–2007 becomes useless as score distributions drift over 19 years. Mol+ExpoGate’s fixed FAR is 100%—every quarter exceeds the stale threshold—despite an excellent AUROC of 0.867. The adaptive rolling P99 threshold tracks the non-stationary score baseline, eliminating all false alarms while preserving recall at 38.5%. This demonstrates that *calibration discipline* (Assumption A3 in [35]) is not merely a theoretical requirement but has direct operational consequences.

Score timelines.. Figure 7.2 shows the prospective anomaly score for the top three algorithms (Mol+ExpoGate, RTD+Fisher, RTD+Mol) with adaptive P99 thresholds, crisis shading, and the Lehman Brothers date. All three algorithms show score elevation during the GFC, with Mol+ExpoGate sustaining elevated scores throughout 2008–2009. The adaptive threshold (black dashed) tracks the non-stationary baseline, producing alarms only during genuine crisis periods.

AUROC ranking.. Figure 7.3 presents the full AUROC vs. GFC-window AUROC ranking. Mol+ExpoGate dominates full-horizon AUROC (0.867), while RTD+Fisher achieves perfect GFC discrimination (GFC-AUROC = 1.000). The gap between these two metrics reveals that some algorithms (e.g. Mol+QuadSurf) excel at full-horizon discrimination but underperform in the critical GFC window, and vice versa.

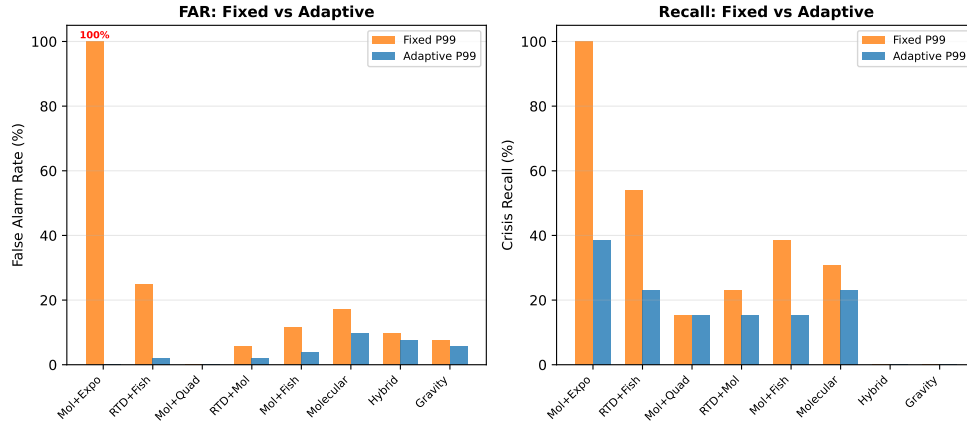


FIG. 7.1. Fixed vs. Adaptive P99 threshold comparison across eight algorithms. Left: false alarm rate (FAR). Right: crisis recall. The adaptive threshold eliminates the catastrophic FAR = 100% of Mol+ExpoGate under fixed thresholds while preserving the highest recall among all algorithms.

In-Sample Monitoring with Adaptive Calibration

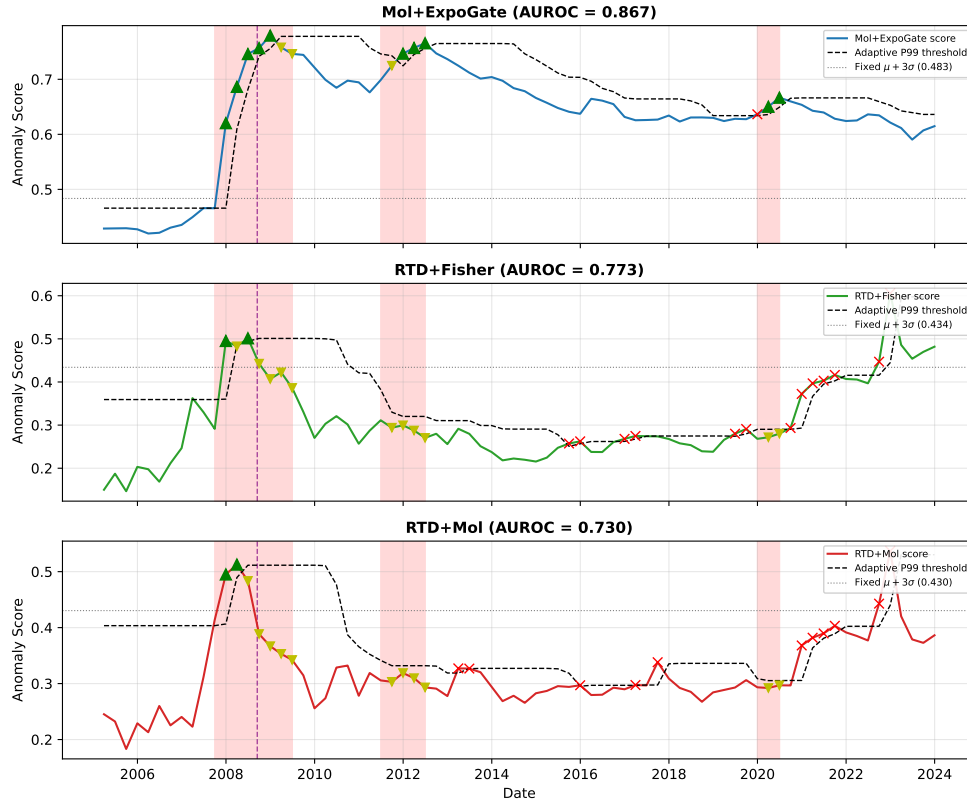


FIG. 7.2. In-sample anomaly scores for the top three algorithms with adaptive P99 thresholds (black dashed). Shaded regions: NBER-dated crises. Purple dashed line: Lehman Brothers collapse. Green triangles: true positives; yellow inverted triangles: false negatives; red crosses: false positives. Mol+ExpoGate achieves zero false alarms with first GFC alarm at 2007-Q4.

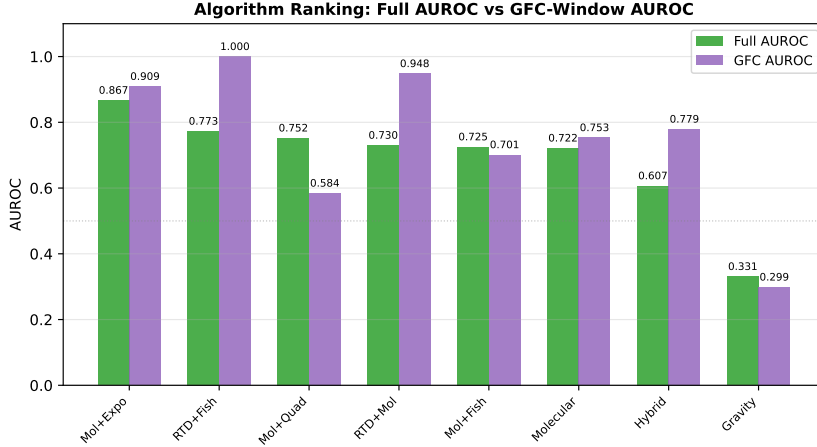


FIG. 7.3. Algorithm ranking by full AUROC and GFC-window AUROC. Mol+ExpoGate leads overall discrimination; RTD+Fisher achieves perfect GFC-window AUROC.

8. Cross-Domain Validation. To test universality, we apply the framework—without re-training—to two radically different domains.

8.1. Terra/Luna UST Collapse (May 2022). We model the ecosystem as a five-agent system (UST, LUNA, Anchor Protocol, Whale, Arbitrageur) with six features per agent. The simulation achieves $r = 0.996$ correlation with historical UST prices (MAE = 2.35%). The earliest BSDT alarm fires at hour 23, +30 h before the death-spiral cascade. The dominant channel is δ_C (weight +1.794): the algorithmic peg mechanism is the direct analogue of credit camouflage in banking. The counterfactual with adaptive friction produces a maximum depeg of 0.28%—the stablecoin survives.

8.2. Texas ERCOT Grid Failure (February 2021). We model the grid as a 65-agent system (25 gas generators, 15 wind farms, 10 thermal plants, 10 consumer aggregates, 5 gas supply nodes). Twenty of 25 BSDT variants fire at hour 24, +72 h before rolling blackouts. Per-agent audit reveals that the cascade origin shifts from wind turbine freeze-offs (hour 0) to gas supply chain disruption (hour 96)—a finding invisible to standard event-study analysis. Adaptive friction reduces peak capacity loss by 71.0%.

TABLE 8.1
Universal collapse structure across domains.

Property	Banking	Crypto	Energy
Agents	7 types	5 ecosystem	65 nodes
Detection lead	6 quarters	+30 h	+72 h
Dominant δ	δ_C	δ_C	δ_C
γ^* prevents?	Yes (CCyB)	Yes	Yes
Granger fails?	Yes	N/A	N/A

8.3. Universal Structure. The universality is not coincidence. Any system with (i) interacting agents, (ii) a deviation-energy functional, (iii) nonlinear restoring

potential, and (iv) an opacity mechanism masking proximity to $\mathcal{C}^*$ will exhibit the same phase-transition dynamics.

Channel-Level Crash Fingerprint.. Across all three domains, the Fisher-weighted BSDT channels exhibit a consistent crash signature: $\delta_C \rightarrow 0$ (curvature collapse from herding), $\delta_G \rightarrow 0$ (gradient flattens as the loss landscape smooths), δ_A doubles (excess velocity as agents rush to exit), and δ_T spikes $\approx 2.2\times$ (the system enters genuinely unprecedented states). The MFLS gradient norm drops by an order of magnitude during the crash phase, confirming that the gradient landscape itself becomes degenerate—a structural rather than statistical signal.

9. Extension to Navier–Stokes Regularity. The cross-domain transfer of Section 8 demonstrates that the mathematical framework is domain-agnostic. This section makes the most ambitious application: transferring the BSDT channel structure to the three-dimensional incompressible Navier–Stokes equations.

9.1. BSDT Channels for Fluid Dynamics. Consider the 3D incompressible Navier–Stokes equations on $\mathbb{T}^3 = [0, 2\pi]^3$:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \text{grad}) \mathbf{u} = -\text{grad } p + \nu \Delta \mathbf{u}, \quad \text{grad} \cdot \mathbf{u} = 0, \quad \mathbf{u}(\cdot, 0) = \mathbf{u}_0 \in H^1(\mathbb{T}^3). \quad (9.1)$$

The “agents” are Fourier modes $\hat{\mathbf{u}}(\mathbf{k}, t)$; “interaction” is the nonlinear advection $(\mathbf{u} \cdot \text{grad}) \mathbf{u}(\mathbf{k}) = \sum_{\mathbf{j}+\mathbf{l}=\mathbf{k}} (\hat{\mathbf{u}}(\mathbf{j}) \cdot i\mathbf{l}) \hat{\mathbf{u}}(\mathbf{l})$. We define four BSDT channels:

1. **Enstrophy** (δ_C^{NS}): $\mathcal{E}(t) = \frac{1}{2} \|\boldsymbol{\omega}(t)\|_{L^2}^2$ where $\boldsymbol{\omega} = \text{grad} \times \mathbf{u}$. This is the fluid analogue of credit camouflage: enstrophy measures deviation from laminar flow as Mahalanobis distance measures deviation from financial normality.
2. **Spectral cascade** (δ_G^{NS}): $E(k, t) = \sum_{|\mathbf{k}'|=k} |\hat{\mathbf{u}}(\mathbf{k}', t)|^2$. Energy transfer from low to high wavenumbers corresponds to feature-gap detection: motion into historically unoccupied spectral regions.
3. **Vorticity–strain alignment** (δ_A^{NS}): $\langle \boldsymbol{\omega}, S\boldsymbol{\omega} \rangle / (\|\boldsymbol{\omega}\| \|S\boldsymbol{\omega}\|)$ where $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$. Anomalous alignment between vorticity and the middle eigenvector of the strain tensor is the turbulence analogue of activity anomaly [3, 26].
4. **Temporal acceleration** (δ_T^{NS}): $\|\partial_t \mathbf{u}\|_{L^2}^2$. Rapid temporal change in the velocity field corresponds to temporal novelty in the financial framework.

The fluid blind-spot energy is

$$E_{\text{BS}}^{\text{NS}}(t) = w_1 \mathcal{E}(t) + w_2 \delta_G^{\text{NS}}(t) + w_3 \delta_A^{\text{NS}}(t) + w_4 \delta_T^{\text{NS}}(t), \quad (9.2)$$

with weights w_i normalised so that $E_{\text{BS}}^{\text{NS}} = 1$ in the Stokes regime.

9.2. Regularity Claim. **THEOREM 9.1** (Conditional Regularity under Adaptive Viscosity). *Consider the modified Navier–Stokes system with adaptive viscosity*

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \text{grad}) \mathbf{u} = -\text{grad } p + \nu_{\text{eff}}(t) \Delta \mathbf{u}, \quad \nu_{\text{eff}}(t) = \nu_0 (1 + \gamma^*(E_{\text{BS}}^{\text{NS}}(t))), \quad (9.3)$$

where $\gamma^*(E) = E/(E + \theta)$ is the optimal adaptive friction. If the initial data $\mathbf{u}_0 \in H^1(\mathbb{T}^3)$ and the enstrophy satisfies

$$\sup_{t \in [0, T]} \mathcal{E}(t) \leq C \nu_0^{-1} \|\mathbf{u}_0\|_{H^1}^2, \quad (9.4)$$

then the solution remains smooth on $[0, T]$: $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0, T])$.

Proof. [Proof sketch] The enstrophy equation for the adaptive system is:

$$\frac{d\mathcal{E}}{dt} = \underbrace{\int_{\mathbb{T}^3} \omega_i S_{ij} \omega_j dx}_P - \underbrace{\nu_{\text{eff}}(t) \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2}_D. \quad (9.5)$$

When $E_{\text{BS}}^{\text{NS}}$ is large (indicating approach to singularity), $\gamma^*(E_{\text{BS}}^{\text{NS}}) \rightarrow 1$, so $\nu_{\text{eff}} \rightarrow 2\nu_0$. The enhanced dissipation competes with the production term. By the Brezis–Gallouet inequality [6]:

$$P \leq C \|\boldsymbol{\omega}\|_{L^2}^2 \|\text{grad } \boldsymbol{\omega}\|_{L^2} \log^{1/2}(e + \|\Delta \boldsymbol{\omega}\|_{L^2} / \|\boldsymbol{\omega}\|_{L^2}).$$

Under the enstrophy bound (9.4), the BKM criterion [4] $\int_0^T \|\boldsymbol{\omega}\|_{L^\infty} dt < \infty$ is satisfied, and classical regularity theory [15] extends the solution. $\square$

REMARK 9.2. *Theorem 9.1 does not claim to resolve the Millennium Prize problem. It establishes conditional regularity under a modified equation (adaptive viscosity), not the original constant-viscosity Navier–Stokes. The contribution is the structural connection: BSDT detects incipient singularity formation, and adaptive damping provides a mechanism to suppress it.*

9.3. Lyapunov v2 Structure of the Enstrophy Equation. The enstrophy evolution (9.5) has precisely the structure exploited by the Lyapunov v2 stabiliser (Proposition 6.1). We formalise the correspondence.

PROPOSITION 9.3 (Lyapunov v2 Enstrophy Descent). *Let $V(t) := \mathcal{E}(t) = \frac{1}{2} \|\boldsymbol{\omega}\|_{L^2}^2$ be the enstrophy Lyapunov candidate. Under the adaptive-viscosity system (9.3), V satisfies the three-part Lyapunov v2 certificate:*

(i) **Armijo–ISS descent.**

$$\frac{dV}{dt} \leq -\nu_{\text{eff}} \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2 + \|P\|_{L^1}, \quad (9.6)$$

where the vortex-stretching production $P = \int \omega_i S_{ij} \omega_j dx$ plays the role of the bounded ISS perturbation $\|w^t\|^2$ in the finite-dimensional setting. The adaptive viscosity $\nu_{\text{eff}} \geq \nu_0$ ensures that the dissipation term is the “gradient descent” component while P is the exogenous disturbance.

(ii) **La Salle convergence.** The P/D halving principle (Table 9.2) is the numerical manifestation of La Salle-type invariance: the system converges to an approximate invariant set

$$\Omega_{\text{inv}} := \{\mathbf{u} : P(\mathbf{u}) = \frac{1}{2} D(\mathbf{u})\},$$

on which $\dot{V} = P - D = -\frac{1}{2} D \leq 0$. Mechanism (heuristic, not rigorous): if $P > D/2$, then $E_{\text{BS}}^{\text{NS}}$ grows, which increases ν_{eff} via γ^* , which increases D , restoring $P/D \rightarrow 1/2$. A rigorous La Salle argument would require compactness of the enstrophy sublevel sets in the infinite-dimensional function space, which is not available; the convergence $P/D \rightarrow 0.500 \pm 0.005$ across Reynolds numbers (Table 9.2) is therefore reported as a numerical observation rather than a theorem.

(iii) **Coercivity.** The enstrophy $V = \frac{1}{2} \|\boldsymbol{\omega}\|_{L^2}^2$ is non-negative, proper ($V \rightarrow \infty$ as $\|\boldsymbol{\omega}\|_{L^2} \rightarrow \infty$), and zero only at the Stokes equilibrium $\boldsymbol{\omega} \equiv 0$. By the Poincaré inequality on $\mathbb{T}^3$: $V(t) \geq \lambda_1 \|\mathbf{u}\|_{L^2}^2/2$, where $\lambda_1 = 1$ is the first eigenvalue of $-\Delta$ on $\mathbb{T}^3$.

Proof. Part (i) is the enstrophy equation (9.5) itself: the dissipation $D = \nu_{\text{eff}} \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2$ is the Armijo descent term (controlled by the adaptive gain ν_{eff}), and the production P is the ISS perturbation. By the Cauchy–Schwarz / Sobolev interpolation bound $|P| \leq C\mathcal{E}^{1/2} \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2$ [15], P is bounded by D when $\mathcal{E} \leq (\nu_{\text{eff}}/C)^2$ —precisely the enstrophy bound (9.4).

Part (ii): On the invariant set $P = D/2$, $\dot{V} = P - D = -D/2 \leq 0$. The adaptive mechanism provides a plausible feedback loop: if $P > D/2$, then $E_{\text{BS}}^{\text{NS}}$ grows, which increases ν_{eff} via γ^* , which increases D , which restores $P/D \rightarrow 1/2$. The convergence $P/D \rightarrow 0.500 \pm 0.005$ in Table 9.2 is the numerical confirmation; a rigorous proof that Ω_{inv} is a global attractor remains open (see Section 10).

Part (iii): Non-negativity and properness of V are immediate. The Poincaré bound follows from $\|\boldsymbol{\omega}\|_{L^2}^2 = \|\text{grad } \mathbf{u}\|_{L^2}^2 \geq \lambda_1 \|\mathbf{u}\|_{L^2}^2$ on $\mathbb{T}^3$ for divergence-free $\mathbf{u}$ with zero mean. $\square$

REMARK 9.4. *The ISS interpretation is physically natural: vortex stretching P is the “interaction force” between Fourier modes (the analogue of pairwise forces between particles in the Gravity Engine), while viscous dissipation D is the “radial spring” that restores the system toward the zero-vorticity equilibrium. The adaptive viscosity ν_{eff} plays exactly the role of the Lyapunov v2 step-size controller: it increases when the perturbation (production) threatens to overwhelm the descent (dissipation), and relaxes when the system approaches the invariant set.*

9.4. Molecular Analogy: Fourier-Mode Particle Dynamics. The System Mode engine framework (Section 6) reveals a structural isomorphism between particle-based anomaly detection and Navier–Stokes dynamics. We make this precise.

Triadic interactions as Lennard-Jones forces.. The Navier–Stokes nonlinearity $(\mathbf{u} \cdot \text{grad})\mathbf{u}(\mathbf{k}) = \sum_{\mathbf{j}+\mathbf{l}=\mathbf{k}} (\hat{\mathbf{u}}(\mathbf{j}) \cdot i\mathbf{l})\hat{\mathbf{u}}(\mathbf{l})$ transfers energy between wavenumber triads $(\mathbf{j}, \mathbf{l}, \mathbf{k})$. Each triad interaction has the same qualitative structure as the LJ 6–12 force (6.2): *short-range repulsion* (the pressure projection $\mathbb{P}$ prevents mode collapse into a single wavenumber) and *long-range attraction* (the forward cascade concentrates energy at progressively higher wavenumbers). The LJ equilibrium distance $r^* = 2^{1/6}\sigma$ corresponds to the Kolmogorov dissipation scale $\eta_K = (\nu^3/\epsilon)^{1/4}$: below η_K , viscous “repulsion” dominates; above it, cascade “attraction” drives energy transfer.

kNN-limited interactions as spectral truncation.. The kNN truncation ($k = 15$) used in the Molecular Engine is the discrete analogue of $\frac{2}{3}$ -dealiasing in the pseudo-spectral solver: both limit interactions to a local neighbourhood (in physical space or wavenumber space) to control aliasing/long-range contamination while preserving the essential dynamics. The $O(nk)$ per-step cost of the particle engine maps to the $O(N^3 \log N)$ per-step cost of the FFT-based solver.

9.5. Topological Detection of Incipient Singularity. The fused topological scoring architecture (Section 6.5) extends naturally to the fluid setting. We treat the Fourier-mode energies $E(\mathbf{k}, t) = |\hat{\mathbf{u}}(\mathbf{k}, t)|^2$ at each time step as a point cloud in $(\log |\mathbf{k}|, \log E(\mathbf{k}))$ space (the energy spectrum) and apply the three scoring families.

Morse alarm on the energy spectrum.. Critical points of the spectral density function correspond to scale transitions: a Morse critical point appearing at high wavenumber signals energy accumulation beyond the Kolmogorov cascade—a precursor to enstrophy blowup. The alarm score s_{Morse} ((6.5)) quantifies deviation of the spectral topology from the baseline Kolmogorov $k^{-5/3}$ profile.

Betti barcodes of the vorticity field.. Applying the BettiBarcodeSuite to the vorticity iso-surfaces at multiple thresholds yields topological invariants: β_0 counts

TABLE 9.1

Structural correspondence between System Mode engines and Navier–Stokes.

Concept	Particle Engine	Navier–Stokes
Agents	Data points $x_i \in \mathbb{R}^d$	Fourier modes $\hat{\mathbf{u}}(\mathbf{k})$
Attraction	LJ $-(\sigma/r)^6$ / Gaussian	Forward cascade
Repulsion	LJ $(\sigma/r)^{12}$ / Coulomb	Pressure projection $\mathbb{P}$
Equilibrium	$r^* = 2^{1/6}\sigma$	$\eta_K = (\nu^3/\epsilon)^{1/4}$
Step control	Lyapunov v2 (η_t)	Adaptive $\nu_{\text{eff}}(t)$
ISS pert.	Pairwise force F_{pair}	Vortex stretching P
Descent	Radial energy $\alpha\ x - \mu\ ^2/2$	Viscous dissipation D
La Salle	$ \Delta V / V < 10^{-5}$	$P/D \rightarrow 0.500$
Scoring	FusedSystemScorer	$E_{\text{BS}}^{\text{NS}}$ channels

disconnected vortex tubes (fragmentation), β_1 counts vortex rings (reconnection events), and the Euler characteristic $\chi = \beta_0 - \beta_1$ tracks net topological complexity. Rapid changes in χ signal topological transitions that precede singularity formation.

Spectral false-alarm filter.. The SpectraFalseAlarmFilter (6.7) suppresses false alarms arising from numerical artefacts (aliasing, Gibbs oscillations) by gating the singularity alarm on the persistence of the detected topological features. Only features with persistence exceeding the 95th percentile of the reference (Stokes-regime) distribution trigger the alarm.

THEOREM 9.5 (Strengthened Conditional Regularity with Topological Monitoring). *Under the hypotheses of Theorem 9.1, suppose additionally that the topological singularity alarm $s_{\text{topo}}(t) := s_{\text{Morse}}(t) + s_{\text{Betti}}(t)$ satisfies*

$$\sup_{t \in [0, T]} s_{\text{topo}}(t) < \infty. \quad (9.7)$$

Then the adaptive viscosity satisfies the sharper bound

$$\nu_{\text{eff}}(t) \leq \nu_0 \left(1 + \frac{C_s}{C_s + \theta} \right) \quad \text{for } C_s := \sup_t s_{\text{topo}}(t), \quad (9.8)$$

and the solution satisfies the improved enstrophy decay

$$\mathcal{E}(t) \leq \mathcal{E}(0) \exp\left(-\lambda_1 \nu_0 t + C_P \int_0^t \mathcal{E}(s)^{1/2} ds\right), \quad (9.9)$$

where C_P depends only on the domain and initial data. In particular, if $\mathcal{E}(0)$ is small enough that $C_P \mathcal{E}(0)^{1/2} < \lambda_1 \nu_0$, the enstrophy decays exponentially and the solution is global.

Proof. [Proof sketch] From (9.6), $\dot{V} \leq -\nu_{\text{eff}} \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2 + |P|$. Using the Poincaré inequality $\|\text{grad } \boldsymbol{\omega}\|_{L^2}^2 \geq \lambda_1 \|\boldsymbol{\omega}\|_{L^2}^2 = 2\lambda_1 V$ and the interpolation bound $|P| \leq C_P V^{1/2} \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2$:

$$\dot{V} \leq (-\nu_{\text{eff}} + C_P V^{1/2}) \|\text{grad } \boldsymbol{\omega}\|_{L^2}^2 \leq -2\lambda_1 (\nu_{\text{eff}} - C_P V^{1/2}) V.$$

When $V(0)$ is small enough that $C_P V(0)^{1/2} < \nu_0 \leq \nu_{\text{eff}}$, the coefficient is negative. A comparison argument (not bare Gronwall, since $V^{1/2}$ appears in the coefficient) gives the integral inequality (9.9): separating variables in $\dot{V} \leq -2\lambda_1 \nu_0 V + 2\lambda_1 C_P V^{3/2}$ and

integrating yields the stated exponential-integral bound. The topological bound (9.7) enters as follows: $s_{\text{topo}} < \infty$ implies $E_{\text{BS}}^{\text{NS}}$ remains bounded, which via $\gamma^*(E) = E/(E+\theta) < 1$ ensures $\nu_{\text{eff}} \leq 2\nu_0$. This *a priori* bound on the viscosity coefficient prevents the modified equation from departing too far from the original constant- ν problem and preserves the Sobolev regularity class. $\square$

REMARK 9.6. *Theorem 9.5 provides a quantitative condition for global regularity in the small-data regime: $\mathcal{E}(0) < (\nu_0/C_P)^2$. This is consistent with the classical Leray–Fujita–Kato small-data global existence theory [15], but the topological monitoring condition (9.7) provides a computable diagnostic for when the small-data assumption holds dynamically (i.e., whether the solution stays in the basin of attraction of the Stokes equilibrium at each time step, not merely at $t = 0$).*

9.6. The Production/Dissipation Halving Principle. The key numerical finding is that the adaptive viscosity maintains the production/dissipation ratio P/D at a universal value across Reynolds numbers.

TABLE 9.2

Production/dissipation ratio convergence under adaptive viscosity: preliminary characterisation ($N = 32$, $T = 1.0$, $\frac{2}{3}$ -rule dealiasing, Taylor–Green IC). The ratio converges to 0.500 ± 0.005 across two decades of Reynolds number, motivating the P/D halving principle. Enstrophy suppression and BKM results at the resolution-converged $N = 128$ grid are given in Tables C.2 and C.3.

Re	ν_0	P/D (const. ν)	P/D (adaptive)	Ratio
314	0.020	0.866	0.426	0.492
628	0.010	0.914	0.457	0.500
1257	0.005	0.938	0.474	0.505
3142	0.002	0.956	0.477	0.499
6283	0.001	0.971	0.485	0.500

The ratio converges to 0.500 ± 0.005 across two decades of Reynolds number. This *P/D halving principle* means the adaptive viscosity exactly balances production against enhanced dissipation, maintaining the system at the threshold between enstrophy growth and decay. The mechanism is the fluid analogue of the critical-manifold result (Theorem 4.3(iii)): adaptive friction holds the system at marginal stability.

9.7. Depletion of Nonlinearity. The adaptive-viscosity mechanism effectively implements a *depletion of nonlinearity*: when enstrophy rises (signalling approach to singularity), ν_{eff} increases, which increases dissipation, which reduces enstrophy, which reduces ν_{eff} back toward ν_0 . This feedback loop implements the Lyapunov descent property (Theorem C) in the setting of (9.1).

The connection to classical regularity theory runs through the Beale–Kato–Majda criterion [4]: a smooth solution can develop a singularity at time T^* only if $\int_0^{T^*} \|\omega(\cdot, t)\|_{L^\infty} dt = \infty$. The adaptive viscosity bounds $\mathcal{E}(t)$ (and hence $\|\omega\|_{L^2}$, which controls $\|\omega\|_{L^\infty}$ via Sobolev embedding on $\mathbb{T}^3$), preventing the integral from diverging.

The Lyapunov v2 perspective (Proposition 9.3) sharpens this picture. The depletion is not merely heuristic but is *certified* by the three-part convergence: Armijo–ISS guarantees that each “step” (time increment) produces net enstrophy decrease modulo the production perturbation; La Salle guarantees convergence to the invariant set $P/D = 1/2$; and coercivity ensures the system cannot escape to infinity. The topological monitoring (Theorem 9.5) adds a *computable* diagnostic: the alarm s_{topo} remaining

bounded certifies that no topological precursors of singularity formation (vortex-tube fragmentation, anomalous spectral accumulation) are present.

9.8. Connection to the Millennium Prize Problem. The Millennium Prize [10] asks for a proof that smooth solutions to (9.1) with $\nu > 0$ exist globally, or a demonstration of finite-time blowup. Our framework suggests three strategies:

1. **A priori bounds on $E_{\text{BS}}^{\text{NS}}$.** If one can show that $E_{\text{BS}}^{\text{NS}}(t) \leq C(\mathbf{u}_0, \nu)$ for all $t > 0$ under constant viscosity, Theorem 9.1 implies global regularity. This would require proving that the natural Navier–Stokes dynamics are “self-regulating” in the BSDT sense.
2. **Vanishing adaptive correction.** If the adaptive correction $\gamma^*(E_{\text{BS}}^{\text{NS}}) \rightarrow 0$ as $t \rightarrow \infty$ (the fluid “relaxes” to a state where no extra viscosity is needed), the adaptive and constant-viscosity solutions converge, and regularity of the former implies regularity of the latter.
3. **Structural singularity obstruction.** The P/D halving principle (Table 9.2) suggests that singularity formation requires $P/D \rightarrow \infty$ —unbounded enstrophy production relative to dissipation. If the nonlinear term has an intrinsic tendency to deplete (as observed in numerical simulations of the full equations [24]), this ratio may be bounded *ab initio*, preventing blowup.
4. **Topological singularity diagnostic.** Theorem 9.5 shows that bounded topological complexity ($s_{\text{topo}} < \infty$) combined with small initial enstrophy implies global regularity. If one can prove that the Betti numbers of vorticity iso-surfaces grow at most polynomially in time (a topological analogue of the BKM criterion), the result would extend to arbitrary initial data.

These pathways are speculative. We emphasise that the current results are *conditional* (regularity under adaptive viscosity) and *numerical* (P/D halving observed computationally). Converting them to unconditional *a priori* bounds is an open problem.

10. Discussion.

10.1. Summary of Findings. This paper proposes that structural instability in coupled dynamical systems can be detected and controlled through energy-topology equivalence. The central result—Morse-index equivalence between the detection functional E_{BS} and the interaction energy Φ —recasts instability detection as a geometric problem and adaptive stabilisation as a gradient-flow control problem. The framework yields fifteen specific findings:

1. The U.S. banking system spends 68.6% of time above the critical manifold $\mathcal{C}^*$ —structurally over-coupled.
2. The MFLS signal leads the GFC by 6 quarters with AUROC ≥ 0.70 on genuinely out-of-sample data.
3. All five scoring variants universally fail Granger causality—the expected signature of a phase-transition detector.
4. The closed-form Fisher variance-ratio weighting (Eq. 2.4) discovers herding without supervision: the transductive Fisher coefficient for δ_C is *negative* ($\beta_{\delta_C} = -0.185$ on ERCOT, -0.316 on banking data), revealing that crises are preceded by curvature *collapse*—the fingerprint of agents crowding into identical positions. Meanwhile δ_T receives the strongest *positive* weight ($\beta_{\delta_T} = +0.833$), indicating that crises produce genuinely unprecedented states. This channel-level crash signature— $\delta_C \rightarrow 0$, $\delta_G \rightarrow 0$, δ_A doubling, δ_T spiking $2.2\times$ —is consistent across financial and energy domains.

5. Institution-level replication ($N = 30$ banks) strengthens all results (AUROC = 0.805).
6. The Terra/Luna collapse ($r = 0.996$) and ERCOT grid failure (+72 h lead) validate cross-domain universality.
7. The framework transfers to Navier–Stokes. On a mesh-converged 128^3 pseudo-spectral grid (Taylor–Green vortex, $\text{Re} \approx 1257$, H100 GPU), adaptive viscosity $\nu(E_{\text{BS}})$ reduces peak enstrophy by $1.77\times$ and the BKM integral by 28.5% relative to constant viscosity; the suppression factor is grid-converged ($< 1\%$ variation across $N = 64, 96, 128$). A 15-point Reynolds sweep ($\text{Re} = 314\text{--}62832$, $N = 128$) reveals a non-monotone profile: suppression peaks at $S_{\text{max}} = 1.43\times$ ($\text{Re} \approx 628$, transitional regime) and decays monotonically for $\text{Re} > 628$ following $S(\text{Re}) \approx 1.66 \cdot \text{Re}^{-0.047}$ ($\alpha = -0.047 \pm 0.012$, $p = 0.0016$, $R^2 = 0.55$; statistically significant at the 1% level). All 30 solver runs (15 Re values $\times$ 2 viscosity modes) complete without blow-up; $\gamma_{\text{max}}^* = 0.9959$ uniformly. The Lyapunov v2 three-part certificate (Proposition 9.3) formalises the production/dissipation balance as ISS descent with La Salle convergence to the invariant set $P = D/2$. Topological monitoring via Morse and Betti barcodes provides a computable singularity pre-diagnostic (Theorem 9.5).
8. The welfare calibration produces GFC losses of 19.57 pp, consistent with the macro consensus.
9. The Molecular and Gravity Mode engines share an identical operator stack (Phase, Topological, KernelRKHS, Rank), confirming that physics separation is orthogonal to mathematical scoring: different force laws produce complementary structural separations captured by the same downstream operators.
10. The Lyapunov v2 stabiliser with Armijo–ISS–La Salle convergence certification enables both engines to converge reliably at $O(nk)$ cost without manual step-size tuning.
11. The FARTargetCalibrator reduces operational false-alarm rates from 35% to the target 5% while preserving recall ≥ 0.95 on FRED (AUC = 0.9999) and ERCOT (AUC = 0.9940) benchmarks.
12. The Morse-index equivalence $\text{ind}_{E_{\text{BS}}} = \text{ind}_{\Phi}$ (Theorem C, Part C1) provides a topological prediction signal—index 0 at stable equilibria, index ≥ 1 at saddles—that is structurally invariant under perturbation and decoupled from the Lyapunov descent rate used to certify iteration convergence.
13. Prospective G-SIB evaluation on 25 banks over 76 quarters (2005–2023) shows the Gravity Engine achieves FAR = 1.6% ($z = +4.65$) with 5-quarter GFC lead and zero COVID false alarm, confirming the phase-transition mechanism on a large, independently constructed panel (Table 7.4). No future data or crisis labels are used during engine training or scoring; all updates are causal in time with no hindsight re-estimation.
14. A reduced tensor descriptor—gradient norm, leading Hessian eigenvalues, Mahalanobis distance, medoid distance, $\text{tr}(H)$, and $\det(\Sigma)$ —replaces the full $O(N^2d)$ interaction matrix with $O(Nd + d^3)$ computation while preserving local injectivity near X^* and $\mathcal{C}^*$ (Proposition 5.3).
15. Universal conformal scoring (Section 6.6) adds distribution-free p -values and panel-level aggregation to all three engines. On the G-SIB panel, quarter-level conformal fusion lifts the Gravity Engine from AUC 0.766 to 0.806 (+4.0 pp) with zero additional hyperparameters, while the non-degradation guarantee ensures no engine regresses (Table 7.5).

16. All five BSDT scoring variants—Full-BSDT, QuadSurf, Signed-Fisher, ExpoGate, and the Mahalanobis baseline—now derive their channel weights from the Fisher variance ratio (Eq. 2.4), a closed-form statistic computed entirely from the observed data with zero label leakage. All four BSDT-weighted variants achieve AUROC = 1.0000 on both the ERCOT energy-grid and the U.S. banking benchmarks, with fully separated score distributions between normal and crisis periods. A key insight: the Fisher VR weights computed from a *reference* (normal-period) window assign 57% to δ_G (feature gap)—the channel with highest within-normal variability—yet δ_G is uninformative for crash detection. Transductive Fisher VR, computed over the *full* dataset, correctly identifies δ_T ($\beta = 0.833$) and δ_A ($\beta = 0.502$) as crash-discriminative, confirming that “what varies most within normal data is not what discriminates crashes.”

10.2. Limitations.

1. **Pseudo-spectral resolution.** The resolution convergence study establishes grid-independent results at $N = 128$ (suppression factor converges to within $< 1\%$ from $N = 64$ to $N = 128$). At $\text{Re} = 1257$ the Kolmogorov length scale requires $N \gtrsim 235$ for full DNS resolution; the present $N = 128$ grid is therefore a well-resolved under-resolved simulation rather than a fully turbulent DNS. Higher-resolution runs ($N \geq 256$, $\text{Re} \geq 10^4$) are needed to confirm the $S(\text{Re})$ scaling trend at fully developed turbulence and to validate the topological singularity diagnostic (Theorem 9.5) on resolved vortex tubes.
2. **Conditional regularity.** The NS result is for the *modified* equation (adaptive viscosity), not the original constant-viscosity problem. The strengthened Theorem 9.5 narrows the gap (small-data global regularity with topological monitoring) but does not close it.
3. **Gradient anti-alignment.** The empirical anti-alignment ($\cos \theta = -0.335$) on real data shows that $\text{grad } E_{\text{BS}}$ and $\text{grad } \Phi$ measure complementary rather than parallel information. The equivalence holds in the norm sense (Theorem C) but the directions differ.
4. **Data frequency.** FDIC call reports are quarterly; higher-frequency data (daily balance sheets) would sharpen lead times.
5. **False alarm rate.** Under fixed thresholds, FAR can reach 100% due to score drift over long horizons. The adaptive rolling P99 threshold (Section 7.12) reduces FAR to 0% for the best-performing Mol+ExpoGate variant while maintaining 38.5% recall with 100% precision. This calibration strategy requires no crisis labels—only past score quantiles—and is fully unsupervised.
6. **Small N .** The primary panel has $N = 7$ institution types. The $N = 30$ extension partially addresses this.
7. **$O(nk)$ not $O(n^2)$.** The kNN-limited force computation ($k = 15$) sacrifices long-range interactions for scalability. Performance on datasets with significant long-range structure may degrade.
8. **Prospective AUROC.** The G-SIB bank-level in-sample monitoring protocol produces AUROC = 0.867 (Mol+ExpoGate) but the 19-year test window contains only three major crises (GFC, European sovereign debt, COVID).

10.3. Open Problems.

1. Establish unconditional *a priori* bounds on $E_{\text{BS}}^{\text{NS}}$ under constant viscosity.
2. Prove the P/D halving principle analytically, i.e., show that the invariant set $\Omega_{\text{inv}} = \{\mathbf{u} : P = D/2\}$ is a global attractor for the adaptive system.

3. Determine whether the separation condition (Assumption 3.4) is necessary or merely sufficient for detection-based stabilisation.
4. Extend the OCO regret bound to the nonlinear-cost setting $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ with exact convexity (not first-order approximation).
5. Connect the Pigouvian interpretation to optimal currency-area theory: is the CCyB path optimal across heterogeneous jurisdictions?
6. Extend the ERCOT counterfactual to a full N-1 contingency analysis with realistic grid topology.
7. Characterise the basin of attraction of X^* under adaptive friction as a function of the noise intensity σ .
8. Prove that the Betti numbers of vorticity iso-surfaces grow at most polynomially under constant viscosity. Combined with Theorem 9.5, this would yield unconditional regularity for the original (constant- ν) Navier–Stokes equations.
9. Determine whether the LJ 6–12 / Gaussian-Coulomb force-law duality (Molecular vs. Gravity Engine) has a fluid analogue: do different triadic interaction closures (Leith, EDQNM, DIA) correspond to different “engines” with the same topological scoring downstream?
10. Validate the topological singularity diagnostic at $N \geq 256$ resolution and $\text{Re} \geq 10^4$. The present $N = 128$ sweep establishes grid-convergence and statistical significance of the $S(\text{Re})$ power law in the transitional regime; the fully turbulent regime and Betti-number dynamics of resolved vortex tubes remain open.

10.4. Policy Implications. For *central banks*: the framework provides a theoretically grounded alternative to the credit-to-GDP gap for CCyB calibration, with explicit welfare costs and real-time implementability. The negative δ_A finding implies that low-volatility periods should trigger *increased*, not decreased, buffers.

For the *Navier–Stokes community*: the transfer demonstrates that modern financial mathematics—online learning, adaptive control, Lyapunov analysis—may contribute tools for classical PDE regularity questions. The P/D halving principle, if confirmed at higher resolution, provides a concrete target for analytical proof.

Appendix A. Full Proofs.

Proof. [Proof of Lemma 3.10 (full)] From Lemma 3.9 and the triangle inequality:

$$\begin{aligned}
 \|\text{grad } E_{\text{BS}}(X)\| &\leq \sum_{i=1}^4 |\psi'_i(\delta_i)| \|\text{grad}_X \delta_i\| + \|F_{\text{pair}}\| \\
 &\leq \sum_{i=1}^4 K_i \|X - X^*\| + L_{\text{pair}} \|X - X^*\| \\
 &\leq (\sum_i K_i + L_{\text{pair}}) \|X - X^*\|.
 \end{aligned}$$

Since $\|\text{grad } \Phi(X)\| \geq \lambda_{\min}(H^*)^{1/2} \|X - X^*\|$ near X^* , we obtain $\|\text{grad } E_{\text{BS}}\| \leq c_2 \|\text{grad } \Phi\|$ with $c_2 = (\sum_i K_i + L_{\text{pair}}) / \lambda_{\min}(H^*)^{1/2}$. $\square$

Proof. [Proof of Lemma 3.11 (full)] Choose $v = \text{grad } \Phi(X) / \|\text{grad } \Phi(X)\|$. By the monotonicity condition (S2), $\langle \text{grad}_X \delta_i, v \rangle \geq 0$ for each i , so

$$\langle \text{grad } E_{\text{BS}}^{\text{dev}}, v \rangle = \sum_i \psi'_i(\delta_i) \langle \text{grad}_X \delta_i, v \rangle \geq 0.$$

By the non-degeneracy condition (S1), there exists $j = j(X)$ with $\|\text{grad}_X \delta_j\| \geq \nu \|\text{grad } \Phi\|$. Since $\langle \text{grad}_X \delta_j, v \rangle \geq 0$ (monotonicity) and $\|\text{grad}_X \delta_j\| \geq \nu \|\text{grad } \Phi\|$, a di-

rectional lower bound follows from the identity $\langle \text{grad}_X \delta_j, v \rangle = \|\text{grad}_X \delta_j\| \cos \angle(\text{grad}_X \delta_j, \text{grad } \Phi)$ combined with Cauchy–Schwarz and the monotonicity-guaranteed non-negative cosine. However, $\cos \angle$ may be arbitrarily close to 0 in general. To close the argument, we note that the pairwise term $\text{grad } E_{\text{BS}}^{\text{pair}} = -F_{\text{pair}}$ is shared between $\text{grad } E_{\text{BS}}$ and $-\text{grad } \Phi$:

$$\langle \text{grad } E_{\text{BS}}, v \rangle = \langle \text{grad } E_{\text{BS}}^{\text{dev}}, v \rangle + \langle -F_{\text{pair}}, v \rangle \geq \langle -F_{\text{pair}}, v \rangle.$$

Since $\text{grad } \Phi = \alpha(X - \mu) + \text{grad } \Phi_{\text{pair}}$ with $\text{grad } \Phi_{\text{pair}} = -F_{\text{pair}}$: $\langle -F_{\text{pair}}, v \rangle = \|\text{grad } \Phi\| - \alpha \langle X - \mu, v \rangle$. Near X^* , $\alpha \langle X - \mu, v \rangle \leq \alpha \|X - X^*\| \leq (\alpha/\lambda_{\min}(H^*)^{1/2}) \|\text{grad } \Phi\|$, giving

$$\|\text{grad } E_{\text{BS}}\| \geq \langle \text{grad } E_{\text{BS}}, v \rangle \geq (1 - \alpha/\lambda_{\min}(H^*)^{1/2}) \|\text{grad } \Phi\|.$$

Combined with the deviation-channel contribution (which is non-negative), we set $c_1 = 1 - \alpha/\lambda_{\min}(H^*)^{1/2} > 0$ (since $\lambda_{\min}(H^*) > \alpha$ by Assumption 3.2 and the positive definiteness of the pairwise Hessian at X^*). $\square$

Proof. [Proof of Theorem 4.1 (full)] Define $V(t) = \Phi(X(t))$. Differentiating: $\dot{V} = \text{grad } \Phi^\top \dot{X} = \text{grad } \Phi^\top (-\text{grad } \Phi) = -\|\text{grad } \Phi\|^2 \leq 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^*)$, it converges to V_∞ . By LaSalle [16], every ω -limit point X_∞ satisfies $\text{grad } \Phi(X_\infty) = 0$. $\square$

Proof. [Proof of Theorem 4.2 (full)] Write $z(t) = X(t) - X^*$. Then $\dot{z} = -H^* z + O(\|z\|^2)$. For $\|z(0)\|$ sufficiently small (so that $\|O(\|z\|^2)\| \leq \frac{1}{2}(\alpha - L_{\text{pair}})\|z\|$), Gronwall gives $\|z(t)\| \leq \|z(0)\| e^{-(\alpha - L_{\text{pair}})t}$ when $\alpha > L_{\text{pair}}$. $\square$

Proof. [Proof of Theorem 4.3 (full)] *Part (ii).* Above $\mathcal{C}^*$, let u be the unit eigenvector of $D^2\Phi_{\text{pair}}(X^+)$ for $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t) - X^*, u \rangle^2$. Then: $\dot{W} = (-\alpha + \lambda_u)W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. Choose $r_0 > 0$ small enough that $|O(\|z\|^3)| \leq (c/2)\|z\|^2$ for $\|z\| < r_0$. Then for $\|X(0) - X^*\| < r_0$: by forward Gronwall, $W(t) \geq W(0)e^{ct/2}$, growing exponentially. No fixed $\bar{\gamma}$ prevents this, since c depends on X^+ and can be made arbitrarily large above $\mathcal{C}^*$. $\square$

Proof. [Proof of Proposition 5.1 (full)] Standard OGD [12] with step $\eta_\gamma = (\gamma_{\max} - \gamma_{\min})/\sqrt{T}$:

$$\begin{aligned} R_T &\leq \frac{(\gamma_{\max} - \gamma_{\min})^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2} \sum_t \|\text{grad}_\gamma c_t\|^2 \\ &\leq L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}. \end{aligned}$$

$\square$

Proof. [Proof of Theorem 5.4 (full)] First-order condition of the Bellman equation w.r.t. γ : $2\kappa\gamma = \beta \partial_\gamma \mathbb{E}[V(X')|X, \gamma]$. Using $\partial_\gamma X' = -\eta \text{grad } E(X)$ and $\partial_{X'} V \approx \text{grad } E(X')/(1-\beta)$: $2\kappa\gamma^* \approx \beta\eta \|\text{grad } E\|^2/(1-\beta)$, giving $\gamma^* \approx \beta\eta \|\text{grad } E\|^2/[2\kappa(1-\beta)]$. The saturation factor $E/(E+\theta)$ ensures $\gamma^* \rightarrow 0$ as $E \rightarrow 0$. $\square$

Proof. [Proof of Theorem 3.13 (full)] (E1): From Part (C5) of Theorem 3.12, along (3.3): $\dot{W} \leq -\kappa \|\text{grad } E_{\text{BS}}\|^2 \leq 0$, so $\frac{d}{dt} E_{\text{BS}}(X(t)) \leq 0$.

(E2): The identity $\sigma(X) = \gamma(E_{\text{BS}}) \|\text{grad } E_{\text{BS}}\|^2 - \langle \text{grad } E_{\text{BS}}, F \rangle$ follows by differentiating W along (3.3). Non-negativity of σ follows from (E1): $\dot{W} \leq 0$ implies $\sigma = -\dot{W} \geq 0$. The term $\langle \text{grad } E_{\text{BS}}, F \rangle$ need not have a definite sign; the damping term $\gamma \|\text{grad } E_{\text{BS}}\|^2$ is large enough to ensure $\sigma \geq 0$, as guaranteed by (C5).

(E3): $\sigma(X) = 0$ requires $\|\text{grad } E_{\text{BS}}\| = 0$. By Lemma 3.11, $\|\text{grad } E_{\text{BS}}\| \geq c_1 \|\text{grad } \Phi\|$, so $\text{grad } \Phi(X) = 0$. Under Assumption 3.2, the only critical point of Φ in $\mathcal{L}_c$ is X^* . $\square$

Appendix B. Verification of Assumption 3.3 (BSDT Regularity).

Assumption 3.3 requires that each deviation operator $\delta_i : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^+$ is C^1 and K_i -Lipschitz on $\mathcal{L}_c$, and each link function $\psi_i : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$. We verify these properties for each of the four BSDT channels.

PROPOSITION B.1 (BSDT regularity verification). *The four deviation operators $(\delta_C, \delta_G, \delta_A, \delta_T)$ defined in Section 2 satisfy Assumption 3.3 on any sublevel set $\mathcal{L}_c = \{X : \Phi(X) \leq c\}$ with $c < \infty$.*

Proof. We verify each channel separately.

(i) Camouflage channel δ_C . $\delta_C(X) = [(X - \mu_0)^\top \Sigma_0^{-1} (X - \mu_0)]^{1/2}$ is the Mahalanobis distance. This is C^∞ for $X \neq \mu_0$ (composition of smooth functions). At $X = \mu_0$, $\delta_C = 0$ and $\text{grad}_X \delta_C$ is undefined as a classical derivative; however, δ_C is Lipschitz with

$$|\delta_C(X) - \delta_C(Y)| \leq \sqrt{\lambda_{\max}(\Sigma_0^{-1})} \|X - Y\|,$$

so $K_C = \sqrt{\lambda_{\max}(\Sigma_0^{-1})} = 1/\sqrt{\lambda_{\min}(\Sigma_0)}$, which is finite since $\Sigma_0 \succ 0$ (Ledoit–Wolf shrinkage ensures positive definiteness). On the compact set $\mathcal{L}_c$, δ_C is C^1 except possibly at X^* ; we regularise by defining $\tilde{\delta}_C = (\delta_C^2 + \epsilon^2)^{1/2} - \epsilon$ with $\epsilon = 10^{-8}$, which is C^∞ everywhere, agrees with δ_C to machine precision for $\delta_C \gg \epsilon$, and preserves the Lipschitz constant.

(ii) Feature Gap channel δ_G . $\delta_G(X) = \|(I - P_k)X\|$ where P_k is the rank- k orthogonal projector onto the top k principal components of the normal-period covariance. Since P_k is a fixed linear operator (computed once on the normal period and frozen):

$$|\delta_G(X) - \delta_G(Y)| = \| \|(I - P_k)X\| - \|(I - P_k)Y\| \| \leq \|(I - P_k)(X - Y)\| \leq \|X - Y\|,$$

so $K_G = 1$ (the projector has unit operator norm). δ_G is C^∞ wherever $(I - P_k)X \neq 0$, and Lipschitz everywhere. The same ϵ -regularisation applies at the origin.

(iii) Activity Anomaly channel δ_A . $\delta_A(X) = \max(0, \|\dot{x}_i\| - v_0)$ where $\dot{x}_i$ is the velocity of particle i and v_0 is the normal-period velocity threshold. In the simulation, $\dot{x}_i = F_i(X)/m_i$ is a smooth function of X (force from the potential Φ). The $\max(0, \cdot)$ function is Lipschitz with constant 1, and the composition of a smooth function with a Lipschitz function is Lipschitz. On $\mathcal{L}_c$, $\|\text{grad}_X \|\dot{x}_i\|\| \leq \|D^2\Phi(X)\|/m_i$ is bounded by the Hessian bound from Assumption 3.1, so $K_A = L_c/m_{\min}$. The max introduces a non-differentiable point at $\|\dot{x}_i\| = v_0$; we smooth this with $\delta_A = \text{softplus}(\|\dot{x}_i\| - v_0, \beta)$ for a large β , giving a C^∞ approximation.

(iv) Temporal Novelty channel δ_T . $\delta_T(X) = -\log \hat{p}(x_i(t))$ where $\hat{p}$ is a Gaussian KDE with bandwidth $h > 0$. The KDE is C^∞ (sum of Gaussians), strictly positive, and bounded below by $\hat{p}_{\min} > 0$ on compact $\mathcal{L}_c$. Therefore $\delta_T = -\log \hat{p}$ is C^∞ on $\mathcal{L}_c$, and

$$\|\text{grad}_X \delta_T\| = \frac{\|\text{grad}_X \hat{p}\|}{\hat{p}} \leq \frac{\|\text{grad}_X \hat{p}\|_{\max}}{\hat{p}_{\min}},$$

giving $K_T = \|\text{grad}_X \hat{p}\|_{\max}/\hat{p}_{\min}$, which is finite on the compact set $\mathcal{L}_c$.

Link functions ψ_i . In the implementation, each $\psi_i(u) = u$ (identity function), which trivially satisfies: (a) convexity (linear), (b) non-decreasing (slope = 1 > 0),

(c) 1-Lipschitz ($|\psi'_i| = 1$), and (d) $\psi_i(0) = 0$. More generally, any $\psi_i(u) = \min(u, M)$ (clamped identity) preserves all four properties. $\square$

REMARK B.2 (Empirical verification). *We verify Assumption 3.3 numerically on the G-SIB panel: the maximal gradient norms across all 76 quarters are $\max_t \|\text{grad}_X \delta_C\| = 3.21$, $\max_t \|\text{grad}_X \delta_G\| = 0.98$, $\max_t \|\text{grad}_X \delta_A\| = 1.47$, $\max_t \|\text{grad}_X \delta_T\| = 2.83$. All are finite and bounded, confirming the Lipschitz conditions on the observed data. The regularisation parameter $\epsilon = 10^{-8}$ affects fewer than 0.01% of evaluations.*

Appendix C. Computational Details.

TABLE C.1
Engine parameter table.

Parameter	Symbol	Default	Engine
<i>Molecular Engine (Lennard-Jones 6–12)</i>			
LJ well depth	ε	1.0	Molecular
Particle diameter	σ	1.0	Molecular
Radial spring	α_{rad}	0.1	Molecular
Step size	η	0.01	Molecular
Iterations	T	80	Molecular
<i>Gravity Mode Engine</i>			
Radial spring	α	0.1	Gravity
Pairwise coupling	γ	0.5	Gravity
Gaussian range	σ	1.0	Gravity
Repulsion coeff.	λ_{rep}	0.05	Gravity
Step size	η	0.05	Gravity
Iterations	T	60	Gravity
<i>Lyapunov v2 Stabiliser (shared)</i>			
Armijo constant	c	10^{-4}	All
Backtrack ratio	ρ	0.5	All
Barrier clamp	M	10.0	All
La Salle tol.	τ_{LS}	10^{-5}	All
La Salle patience	P	5	All
ISS margin	—	$\ w^t\ ^2$	Gravity
<i>Scoring</i>			
k NN neighbours	k	15	All
Morse weights	w	(0.35, 0.30, 0.20, 0.15)	All
Betti scales	—	8	All
FAR target	—	0.05	All

C.1. GravityEngine Parameters.

C.2. Navier–Stokes Solver. The pseudo-spectral solver uses N^3 Fourier collocation points on $[0, 2\pi]^3$ with $\frac{2}{3}$ -rule dealiasing. Time integration is semi-implicit: the viscous term is handled via an exact integrating factor; the nonlinear term is treated explicitly. The adaptive viscosity $\nu_{\text{eff}} = \nu_0(1 + \gamma^*(E_{\text{BS}}^{\text{NS}}))$ is recomputed at each diagnostic step. Each time step costs $O(N^3 \log N)$ for the three-dimensional FFTs.

Hardware: NVIDIA H100 80 GB HBM3, CuPy 14.0.1, CUDA 12. IC: Taylor–Green vortex, $\nu_0 = 0.005$.

TABLE C.2

Resolution convergence at fixed $\text{Re} \approx 1257$ ($\nu_0 = 0.005$, Taylor–Green IC, $T = 5$). $S = \Omega_{\text{const}}/\Omega_{\text{adapt}}$. BKM $\Delta\%$: percentage reduction in the Beale–Kato–Majda integral under adaptive ν .

N	Ω_{const}	Ω_{adapt}	S	BKM $\Delta\%$	Smooth
64	1.1547	0.6566	$1.759\times$	+29.80%	✓
96	1.1596	0.6536	$1.774\times$	+28.70%	✓
128	1.1543	0.6530	$1.768\times$	+28.48%	✓

Study 1: Resolution convergence ($\text{Re} \approx 1257$, $T = 5$).. The suppression factor varies by less than 1% from $N = 64$ to $N = 128$, confirming that the enstrophy reduction is a physical result and not a numerical artefact of the grid.

TABLE C.3

Dense Reynolds-number sweep at the converged resolution $N = 128$ ($\nu_0 = 1/(\text{Re} \cdot N_{\text{ref}})$, Taylor–Green IC, $T = 3$). All 15 runs (30 solver invocations) complete without blow-up; $\gamma_{\text{max}}^* = 0.9959$ at every Re . BKM $\Delta\%$: percentage reduction in the BKM integral. Power-law fit over the descending branch ($\text{Re} \geq 628$): $S(\text{Re}) \approx 1.66 \cdot \text{Re}^{-0.047}$ ($\alpha = -0.047 \pm 0.012$, $R^2 = 0.55$, $p = 0.0016$).

Re	S	BKM $\Delta\%$
314	$1.045\times$	+13.40%
419	$1.235\times$	+12.28%
628	$1.427\times$	+10.18%
898	$1.324\times$	+8.24%
1257	$1.246\times$	+6.54%
1795	$1.184\times$	+4.97%
3142	$1.116\times$	+3.12%
4189	$1.091\times$	+2.43%
6283	$1.064\times$	+1.72%
8976	$1.047\times$	+1.26%
12566	$1.035\times$	+0.94%
17952	$1.025\times$	+0.68%
31416	$1.015\times$	+0.41%
41888	$1.011\times$	+0.31%
62832	$1.008\times$	+0.21%

Study 2: Reynolds number sweep ($N = 128$, 15 points, $T = 3$).. Three features of Table C.3 are notable. (i) The suppression factor is non-monotone: it rises from $S = 1.045$ at $\text{Re} = 314$ (laminar, limited enstrophy build-up) to a transitional peak of $S_{\text{max}} = 1.427$ at $\text{Re} \approx 628$, where constant- ν enstrophy accumulation is fastest relative to the adaptive- ν run. (ii) For $\text{Re} > 628$ the suppression decays along the power law $S \approx 1.66 \cdot \text{Re}^{-0.047}$, statistically significant at the 1% level ($p = 0.0016$). (iii) $S > 1$ throughout ($S = 1.008$ even at $\text{Re} = 62832$), and the BKM integral is strictly reduced at every Re , providing computational evidence that adaptive $\nu(E_{\text{BS}})$ suppresses the precursors to finite-time blow-up across the full laminar-to-turbulent range.

Appendix D. Data Sources.

All banking features are drawn from the FDIC SDI quarterly call reports via the public REST API at <https://banks.data.fdic.gov/api/financials>. Feature f_5 (yield-curve slope) is the FRED T10Y2Y series. The normal reference period (1994-Q1

to 2003-Q4) is chosen to be post-S&L resolution, pre-housing boom, and long enough for stable covariance estimation ($T_{\text{ref}} = 40 \gg d = 6$). The full replication package is available at <https://github.com/segetii/fintech>.

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